Dynamic Matrix Recovery

Ziyuan Chen
School of Mathematical Sciences, Center for Statistical Science,
Peking University, Beijing, China
Ying Yang
Academy of Mathematics and Systems Science,
Chinese Academy of Sciences, Beijing, China
Fang Yao *
School of Mathematical Sciences, Center for Statistical Science,
Peking University, Beijing, China

Abstract

Matrix recovery from sparse observations is an extensively studied topic emerging in various applications, such as recommendation system and signal processing, which includes the matrix completion and compressed sensing models as special cases. In this work, we propose a general framework for dynamic matrix recovery of low-rank matrices that evolve smoothly over time. We start from the setting that the observations are independent across time, then extend to the setting that both the design matrix and noise possess certain temporal correlation via modified concentration inequalities. By pooling neighboring observations, we obtain sharp estimation error bounds of both settings, showing the influence of the underlying smoothness, the dependence and effective samples. We propose a dynamic fast iterative shrinkage-thresholding algorithm that is computationally efficient, and characterize the interplay between algorithmic and statistical convergence. Simulated and real data examples are provided to support such findings.

Keywords: compressed sensing, local smoothing, low rank, matrix completion

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1 Introduction

Matrix recovery from sparse observations has been extensively studied in a wide range of problems, such as signal processing (Li et al. 2019), recommendation system (Koren et al. 2009) and the quantum state tomography (Gross et al. 2010). The classical model is to link the response Y with the random matrix $X \in \mathbb{R}^{m_1 \times m_2}$ via

$$Y = \mathrm{Tr}(X^\top M^0) + \xi \tag{1}$$

to recover the low-rank matrix $M^0 \in \mathbb{R}^{m_1 \times m_2}$ from the independent observations $\{(X_i, Y_i)\}_{i=1}^n$, where $\mathrm{Tr}(\cdot)$ is the trace of a matrix, ξ is a mean zero random error and n denotes the sample size, which is typically much smaller than $m_1 m_2$ (i.e., $n \ll m_1 m_2$). By imposing different assumptions on X, the model includes a variety of matrix recovery models, in which matrix completion and compressed sensing are two important cases. Most existing works assume that the target matrix M^0 is static (Klopp 2011, Koltchinskii & Xia 2015, Fan et al. 2021, Xia & Yuan 2021). However, there are various scenarios where M^0 is dynamic and observed at multiple time points, such as in time-varying user interests in recommendation systems (Xu et al. 2020), temporally-changed graph models (Qiu et al. 2017) and dynamic quantum dot matrices (Csurgay et al. 2000). To formulate a general framework accounting the dynamic changes in numerous applications, we propose investigating the recovery of the dynamic matrix as follows,

$$Y_t = \mathrm{Tr}(X_t^\top M_t^0) + \xi_t,\ t = 1, \ldots, T, \tag{2}$$

where t is the timestamp, $\{Y_t, X_t, \xi_t\}$ is the tuple of the response, the design matrix and the noise at time t, respectively, and M_t^0 is a dynamic matrix which is assumed smooth across time t with a low-rank structure.

A prevalent line of research for solving the classical static model (1) is to impose strict

low-rank constraints based on matrix factorization. For instance, Keshavan et al. (2009) proposed singular value decomposition to make the recovery, Zhao et al. (2015) used QR decomposition instead, and Zheng & Lafferty (2016) adopted Burer-Monteiro factorization for rectangle matrix completion. Another stream is the convex approaches, thanks to Candès & Recht (2009) that first relaxed the non-convex rank restriction to nuclear norm for efficient optimization. A variety of convex algorithms and theoretical results are further established (Recht et al. 2010, Candes & Plan 2011, Agarwal et al. 2012). Though M_t^0 in (2) can be recovered by adopting the static matrix recovery at each time t, these *single-stage* estimates are less desirable, as they only use data at time t to reconstruct M_t^0 and fail to take advantage of the neighboring observations and smoothness of the underlying matrix. Related work also includes tensor regression to rebuild M_t^0, which views the dynamic model (2) from a static perspective by letting $N \in \mathbb{R}^{T\times m_1\times m_2}$ be a three order tensor whose (t,i,j)th element is the (i,j)th element of M_t^0. Gandy et al. (2011) and Liu et al. (2012) proposed algorithms for tensor completion and Zhang et al. (2020) considered the low-rank tensor trace regression via the importance sketching algorithm. We mention that their low-rank assumption of N is different from model (2) that M_t^0 is dynamically and smoothly evolving along time t. Consequently, one needs to process the whole data to recover M_t in tensor regression setting, which is time- and space-consuming.

A more efficient approach is to adopt the dynamic modeling/algorithm that avoids computing the whole data for update and has been adopted in some relevant problems. Lois & Vaswani (2015) designed a memory-saving algorithm for batch robust principle component analysis; Gao & Wang (2018) studied dynamic robust principal component analysis and proposed a fast iterative algorithm; Hu & Yao (2022) proposed a unified framework to directly estimate dynamic principal subspace in high dimension. The dynamic

algorithm for matrix recovery has been explored in Xu & Davenport (2016), which assumed that ξ_t are independently and identically distributed (i.i.d.) zero-mean Gaussian error and imposed a parametric assumption on structure of the underlying matrix, $M_t^0 = M_{t-1}^0 + \varepsilon_t$, where ε_t are i.i.d. zero-mean Gaussian noises. Another line of work, namely the dynamic tensors, also finds extensive applications in various domains, and several works have been noted in this direction. For instance, Zhang et al. (2021) proposed a time-varying coefficient model for temporal tensor factorization, employing a polynomial spline approximation, and Zhou et al. (2013) introduced a family of rank-R generalized linear tensor regression models. Most of the current research on dynamic tensors relies on tensor decomposition techniques and assumes that only certain decomposition factors or coefficients are changing (Zhang et al. 2021, Bi et al. 2021, Koren 2009, Wang et al. 2016).

Different from the existing work, we consider the general scenario where the dynamics are not limited to decomposition components or a parametric structure *priori*. There exist two major challenges. First, one needs to maintain the low-rank structure in the whole procedure of dynamic algorithm. Second, the combination of neighboring information often inflates the estimation error when the observations are correlated along time. To resolve these, we utilize local smoothing which has the advantages that the manifold structure of low-dimensional matrices can be approximately retained and temporal dependence has less influence when T increases. Based on these considerations, we propose a dynamic matrix recovery algorithm without imposing structural assumption besides smoothness and allowing the measurement error ξ_t and/or design matrix X_t to be possibly dependent across time. The idea is to pool together the observations that lie in the local window to estimate M_t^0. Noting that M_t^0 is low-rank, we set the target function composed of a locally weighted loss function and the nuclear norm penalty of M_t^0 and propose a Dynamic Fast

Iterative Shrinkage-Thresholding Algorithm (DFISTA) for optimization.

The main contributions of this work are summarized as follows. First, we establish a general framework for dynamic matrix recovery with theoretical guarantees. We allow the underlying matrix varying smoothly under the low-rank constraint and allow the observations to be dependent across time, which has more flexibility and applicability. While existing works only study the estimation error bounds for specific problems such as matrix completion and compressed sensing (Koltchinskii et al. 2011a, Negahban & Wainwright 2011, Xu & Davenport 2016, Fan et al. 2021), we derive the error bounds under this general framework, which can be readily adopted to such matrix recovery problems. Second, the proposed method attains a faster convergence rate than classical static methods in the following sense. Take the matrix completion for instance. The ratio of error bounds of the proposed estimator compared to the classical one (i.e., single-stage estimates at each t) is $O_p\big(\{m \log m/n_t\}^{-1/10} T^{-2/5}\big)$, where n_t is the number of observations at time t and $m = m_1 \vee m_2$ ("$\vee$" indicates the greater of two quantities). When the observations are sparse in the sense that $n_t \ll m \log m$, the ratio becomes $o_p\big(T^{-2/5}\big)$. Specifically, the dynamic method uses only $n' = \big(m \log m/n\big)^{-1/4} n/T = o(n/T)$ observations to attain the same convergence rate as the static method using n observations. Last but not least, a computationally efficient algorithm is devised for dynamic matrix recovery. This algorithm iterates only a subset of the data to update the estimate at each step. By using the estimate of the last step as the initial of the current optimization procedure, the algorithm converges faster than the single-stage recovery. Specifically, the ratio of computational complexities of our algorithm compared to the single-stage recovery is $O(T^{-3/5} \vee n^{-1/2})$.

The rest of the paper is organized as follows. In Section 2, we present the proposed method and algorithm for dynamic matrix recovery, and establish theoretical guarantees

in Section 3 for both independent and dependent observations with algorithmic analysis. Finally, we conduct simulation and real data experiments in Section 4 and 5 to show the favorable performance of the proposed dynamic method/algorithm compared with other existing methods. Additional theoretical results and technical proofs are deferred to an online Supplementary Material due for space economy.

2 Methodology and Algorithm

We consider the general framework in the form of dynamic trace regression whose coefficient matrix is changing smoothly along time. Specifically, at time $t = 1, \ldots, T$, we observe (Y_{ti}, X_{ti}) following model (2),

$$Y_{ti} = \mathrm{Tr}(X_{ti}^\top M_t^0) + \xi_{ti}, \quad i = 1, \ldots, n_t,$$

where M_t^0 is the time-varying coefficient matrix, X_{ti} and ξ_{ti} are design matrices and zero-mean noises, respectively, $\{X_{ti}, \xi_{ti}\}_i$ (given t) are mutually independent across subject and $\{X_{ti}, \xi_{ti}\}_t$ (given i) may be correlated across time. Parallel to the static model (1), the dynamic trace regression includes several problems such as dynamic matrix completion and dynamic compressed sensing that can be applied to dynamic recommendation systems and video signal processing as illustrated in numerical experiments of Section 5.

Example 1 *(Dynamic matrix completion). For each t, suppose that the design matrices X_{ti} are i.i.d. uniformly distributed on the set*

$$\mathcal{E} = \{e_j(m_1)e_k^\top(m_2) : 1 \le j \le m_1, 1 \le k \le m_2\}, \tag{3}$$

where $e_j(m)$ are the canonical basis vectors in $\mathbb{R}^m$ and then $\mathcal{E}$ forms an orthonormal basis in the space $\mathbb{R}^{m_1 \times m_2}$. Then estimating M_t^0 is equivalent to the problem of matrix completion under uniform sampling at random (USR).

Example 2 *(Dynamic compressed sensing). For each t, suppose that the design matrices X_{ti} are i.i.d. replicates of a random matrix X_t such that $\langle M, X_t\rangle$ is a sub-gaussian random variable for any $M \in \mathbb{R}^{m_1\times m_2}$. In this work, we focus on the concrete example that each design matrix X_{ti} is a random matrix whose elements are independent mean-zero sub-gaussian random variables with variance $\sigma^2_{X_t}$. Moreover, if the design X_t is homogeneous across t, we denote the variance as σ^2_X.*

We first briefly review the classical static method in the context of dynamic modeling, i.e., adopting the static method at each time t, which is referred to as the single-stage estimates in the sequel. Define the matrix inner product by $\langle M, X\rangle = \mathrm{Tr}(M^\top X)$. Koltchinskii et al. (2011a) proposed to incorporate the knowledge of the distribution of X_{ti} and estimate M_t^0 by

$$\widehat{M}_t^\lambda = \underset{M\in\mathbb{M}}{\mathrm{argmin}} \left[\frac{1}{n_t}\sum_{i=1}^{n_t} \mathbb{E}(\langle M, X_{ti}\rangle)^2 - \left\langle \frac{2}{n_t}\sum_{i=1}^{n_t} Y_{ti}X_{ti}, M \right\rangle + \lambda\|M\|_1 \right], \tag{4}$$

where $\mathbb{M} \subset \mathbb{R}^{m_1\times m_2}$ is a convex set of matrices, $\lambda > 0$ is a regularization parameter and $\|M\|_1$ is the nuclear norm of M. They proved that in the setting of matrix completion in Example 1, if λ is appropriately set, with probability at least $1 - 3/(m_1 + m_2)$,

$$(m_1m_2)^{-1/2}\|\widehat{M}_t^\lambda - M_t^0\|_2 \le C\left\{\frac{\log(m_1+m_2)\max(m_1,m_2)r_t}{n_t}\right\}^{1/2},$$

where $r_t = \mathrm{rank}(M_t^0)$ and $\|M\|_2$ is the Frobenius norm of matrix M. Though they have proved that the rate is optimal up to logarithmic factors comparing to the minimax lower bounds in classical trace regression, this approach fails to incorporate temporal information and hence is not efficient in multi-stage case.

A direct modification is to utilize the *two-step* procedure that first obtains the single-stage estimates $\widehat{M}_t^\lambda$ and then applies local smoothing to $\{\widehat{M}_t^\lambda\}_t$ to achieve the final estimate

as follows,

$$\widehat{M}_t^{\text{t-s}} = \sum_{j=1}^{T} \omega_h(j-t)\widehat{M}_j^{\lambda}, \tag{5}$$

where the local smoothing weights $\omega_h(j-t) = K_{Th}(j-t) / \sum_{k=1}^{T} K_{Th}(k-t)$ with the kernel function $K(x)$ and bandwidth h. The two-step estimation approach computes a weighted average of low-rank matrices that have been estimated at distinct time points. However, this method encounters a significant challenge when dealing with sparse observations. In such scenarios, the estimated matrix for each time point could potentially disrupt the true low-rank structure, displaying significant deviation from the actual value. Unfortunately, this disparity cannot be corrected through smoothing techniques.

Motivated by the observation that the manifold structure of low-rank matrices can be approximately preserved locally without matrix factorization, we design the dynamic target function with nuclear norm penalty, i.e.,

$$\widetilde{M}_t^{\lambda} = \underset{M\in\mathbb{M}}{\operatorname{argmin}} \sum_{j=1}^{T} \omega_h(j-t) \left[\frac{1}{n_j} \sum_{i=1}^{n_j} \mathbb{E}(\langle M, X_{ji}\rangle)^2 - 2\left\langle \frac{1}{n_j}\sum_{i=1}^{n_j} Y_{ji}X_{ji}, M \right\rangle \right] + \lambda\|M\|_1, \tag{6}$$

which combines the observations across time and employs local smoothing for incorporating the temporal information. It is worth noting that in many applications including our Examples 1 and 2, the distribution of X_t is known, and yet this information has not been exploited in empirical risk. Thus we may utilize the distribution knowledge by substituting the empirical form with true values $\mathbb{E}\langle M, X_{ji}\rangle$. The temporal smoothness of M_t^0 guarantees that information in adjacent time points can be utilized to reduce the estimation error. We mention that the classical estimate $\widehat{M}_t^{\lambda}$ in (4) can be regarded as a special case of the proposed local estimate $\widetilde{M}_t^{\lambda}$ in (6) as the bandwidth $h \to 0$. By choosing h appropriately, our estimate pools more observations to reduce the estimation variance while controlling the bias caused by the varying of M_t^0, as discussed in detail in Section 3.

For implementation, we adopt the Fast Iterative Shrinkage-Thresholding Algorithm

(FISTA) which is an accelerated proximal method proposed by Beck & Teboulle (2009). It improves traditional proximal methods and terminates in $\mathcal{O}(\varepsilon^{-1/2})$ iterations with a ε-optimal solution. To handle with the trace norm penalty optimization problem (6), we use the matrix form FISTA established by Ji & Ye (2009) and Toh & Yun (2010). Define the objective function in (6) as $F_t(M) = f_t(M) + g(M)$, where

$$f_t(M) = \sum_{j=1}^{T} \omega_h(j-t) \left[\frac{1}{n_j} \sum_{i=1}^{n_j} \mathbb{E}(\langle M, X_{ji} \rangle)^2 - 2 \left\langle \frac{1}{n_j} \sum_{i=1}^{n_j} Y_{ji} X_{ji}, M \right\rangle \right].$$

We mention that the gradient function of f_t is equipped with Lipschitz constant L_f presented in Section 3.4, and that $g(M) = \lambda \|M\|_1$ is a continuous convex function. Then the matrix form FISTA can be applied to solve the dynamic trace regression problem

$$\widetilde{M}_t^{\lambda} = \arg\min_{M \in \mathbb{M}} f_t(M) + g(M).$$

Note that $\widetilde{M}_t^{\lambda}$ need to be calculated for a sequence $t = 1, 2, \cdots, T$. The algorithm is accelerated at time t by utilizing the estimate from the timestamp $(t-1)$. Specifically, suppose that at time t we apply the matrix form FISTA for k_t iteration times to numerically approximate the estimate $\widetilde{M}_t^{\lambda} = \arg\min_{M \in \mathbb{M}} f_t(M) + g(M)$ denoted by $M_t^{(k_t)}$. Then input $M_t^{(k_t)}$ as the initial matrix in the procedure of estimating $\widetilde{M}_{t+1}^{\lambda} = \arg\min_{M \in \mathbb{M}} f_{t+1}(M) + g(M)$ at time $(t+1)$. Such initialization method accelerates the algorithm, as demonstrated in Section 3.4. We summarize the implementations of the proposed DFISTA in Algorithm 1. In the algorithm, "svd" means calculating the single value decomposition components U, D, V for a matrix $G \in \mathbb{R}^{m_1 \times m_2}$ and "$(\cdot)_+$" means the threshold operator such as $(D - \gamma)_+ = \mathrm{diag}((\lambda_1 - \gamma) \vee 0, \cdots, (\lambda_{m_1} - \gamma) \vee 0)$ where $D = \mathrm{diag}(\lambda_1, \cdots, \lambda_{m_1})$. The learning rate s_k is updated according to Nesterov (1983).

Algorithm 1 Dynamic Low-Rank Matrix Recovery

Input: $\sum_{t=1}^{T} n_t$ matrices $X_{ti} \in \mathbb{R}^{m_1 \times m_2}$ and $\sum_{t=1}^{T} n_t$ scalars Y_{ti}, $t = 1, 2, \ldots, T$, $i = 1, 2, \cdots, n_j$, max iteration steps K and torrent `tor`.

Output: T matrices $M_t \in \mathbb{R}^{m_1 \times m_2},\ t = 1, 2, \ldots, T$.

Choose a initial matrix $M_0 \in \mathbb{R}^{m_1 \times m_2}$.

while $t \leq T$ **do**

  $M_t^{(-1)} \leftarrow M_{t-1},\ M_t^{(0)} \leftarrow M_{t-1}$;

  $s_0 \leftarrow 1,\ s_{-1} \leftarrow 1$;

  for $k = 0, 1, \ldots, K$ **do**

    $N_t^{(k)} \leftarrow M_t^{(k)} + \frac{s_{k-1}-1}{s_k}\left(M_t^{(k)} - M_t^{(k-1)}\right)$;

    $G_t^{(k)} \leftarrow N_t^{(k)} - L_f^{-1} \sum_{j=(t-Th/2)\vee 0}^{(t+Th/2)\wedge T} \frac{w_h(j-t)}{n_j} \sum_{i=1}^{n_j} X_{ji}\left(\langle X_{ji}, N_t^{(k)}\rangle - Y_j\right)$;

    $U, D, V \leftarrow \mathtt{svd}(G_t^{(k)})$;

    $M_t^{(k+1)} \leftarrow U(D - 2\lambda/L_f)_+ V^T$;

    $s_{k+1} \leftarrow \frac{1+\sqrt{1+4s_k^2}}{2}$;

    if $|F_t(M_t^{(k+1)}) - F_t(M_t^{(k)})| \leq \mathtt{tor}$ **then**

      break

    end if

  end for

  $M_t = M_t^{(k+1)}$;

end while

return $M_t, t = 1, 2, \cdots, T$

Unlike most factorization based algorithms that are challenging to analyze for their convergence properties, we are able to analyze the error bounds of the algorithm's results with respect to the true value. This analysis bridges the gap between the statistical estimator and the estimator obtained from the algorithm, as discussed in detail in Section 3.4.

3 Theoretical Guarantees

In this section, we study both the statistical properties of the estimation error $\|\widetilde{M}_t^{\lambda} - M_t^0\|_2$ and the algorithmic convergence of the proposed method. We first present the error bound of the estimate $\widetilde{M}_t^{\lambda}$ for general dynamic trace regressions. Then we derive the explicit expressions of the upper bound of $\|\widetilde{M}_t^{\lambda} - M_t^0\|_2$ when $\{X_t\}_t$ and $\{\xi_t\}_t$ are independent across t and when $\{X_t\}_t$ and $\{\xi_t\}_t$ are ϕ-mixing processes, respectively. Noting that X_t and/or ξ_t may be unbounded, the assumptions for classical concentration inequalities are violated (Merlevède et al. 2009, 2011, Hang & Steinwart 2017). We thus modify the Talagrand's inequalities and adopt truncation technique appropriately to derive the convergence rate for the dependent setting. In both scenarios, we shall explain how the number of time points T, the smoothness of M_t^0 and the dependence of X_t and ξ_t across time influence the estimation error explicitly.

In the sequel, we denote $\|M\|_0$, $\|M\|_1$, $\|M\|_2$, $\|M\|_\infty$ and $\langle M, N\rangle$ as the rank, nuclear norm, Frobenius norm, maximum singular value of M and the trace of $M^\top N$, respectively.

3.1 General error bounds

Let $\mathrm{vec}(X) \in \mathbb{R}^{m_1 m_2}$ be the vectorization of $X \in \mathbb{R}^{m_1 \times m_2}$ and define the second moment matrix of X_t by $\Sigma_t \in \mathbb{S}^{m_1 m_2 \times m_1 m_2}$ such that

$$\Sigma_t = \frac{1}{n_t} \sum_{i=1}^{n_t} \mathbb{E}[\mathrm{vec}(X_{ti})\,\mathrm{vec}(X_{ti})^\top].$$

It is necesary to guarantee the identifiability of matrix recovery that the positive semi-definite matrix Σ_t is invertible. To simplify, we consider the homogeneous scenario such that at each time point t, the design $\{X_t\}_t$ have the same second moment matrix as stated below.

Assumption 1 *There exists a positive definite matrix* $\Sigma \in \mathbb{S}_+^{m_1 m_2 \times m_1 m_2}$ *such that* $\Sigma_j = \Sigma,\ j = 1, 2, \ldots, T$ *with smallest eigenvalue* $\mu > 0$ *and bounded condition number* κ_Σ.

We remark that in the heterogeneous situation, the convergence rate can be derived similarly, which is at the same order as the homogeneous setting with a more tedious presentation. An important indication of Assumption 1 is that, for all $M \in \mathbb{R}^{m_1 \times m_2}$ and $j = 1, 2, \cdots, T$,

$$\frac{1}{n_j} \sum_{i=1}^{n_j} \mathbb{E}(\langle M, X_{ji} \rangle)^2 = \operatorname{vec}(M)^\top \Sigma \operatorname{vec}(M) \geq \mu \|M\|_2^2,$$

which is a standard assumption in matrix recovery. The interpretation is that the smallest eigenvalue $\mu > 0$ determines the information contained in the design matrix X_t. Intuitively, the larger μ is, the more information is revealed by data, thus the more precise the estimator $\widetilde{M}_t^\lambda$ is. In most cases, μ can be calculated directly. For instances, in matrix completion of Example 1, the smallest eigenvalue of Σ is $\mu = 1/m_1 m_2$, and in compressed sensing of Example 2, μ is the smallest variance of elements in X_{ji}, which is equal to σ_X^2 under Assumption 1.

For brevity, we further denote $\Delta_j = 1/n_j \sum_{i=1}^{n_j} (Y_{ji} X_{ji} - \mathbb{E}(Y_{ji} X_{ji}))$ and

$$\delta_h M_t = \left\| M_t^0 - \sum_{j=1}^{T} \omega_h(j-t) M_j^0 \right\|_2, \quad \mathcal{W}_h \Delta_t = \left\| \sum_{j=1}^{T} \omega_h(j-t) \Delta_j \right\|_\infty.$$

The following theorem gives the general upper bound of estimation error $\|\widetilde{M}_t^\lambda - M_t^0\|_2$.

Theorem 1 *Under Assumption 1, if* $\lambda \geq 2\mathcal{W}_h \Delta_t$, *then*

$$\left\| \widetilde{M}_t^\lambda - M_t^0 \right\|_2 \leq \delta_h M_t + \min \left\{ \left((\delta_h M_t)^2 + \frac{2\lambda}{\mu} \left\| M_t^0 \right\|_1 \right)^{1/2}, \frac{1+\sqrt{2}}{2} \frac{\lambda}{\mu} \sqrt{r_t} \right\}. \tag{7}$$

When selecting $\lambda = 2\mathcal{W}_h \Delta_t$, *we have*

$$\left\| \widetilde{M}_t^\lambda - M_t^0 \right\|_2 \leq \delta_h M_t + \min \left\{ \left((\delta_h M_t)^2 + 2\mu^{-1} \mathcal{W}_h \Delta_t \left\| M_t^0 \right\|_1 \right)^{1/2}, (1+\sqrt{2}) \mu^{-1} \mathcal{W}_h \Delta_t \sqrt{r_t} \right\}. \tag{8}$$

The term $\delta_h M_t$ corresponds to the bias caused by kernel smoothing which is affected by kernel $K(x)$ and the bandwidth h, and $\mathcal{W}_h\Delta_t$ corresponds to the variance term caused by the effective samples in the local window and the measurement error ξ. When $h = 0$, the result in Theorem 1 coincides with the classical result of Corollary 1 in Koltchinskii et al. (2011a) such that when $\lambda = 2\|\Delta_t\|_\infty$,

$$\left\|\widehat{M}_t^\lambda - M_t^0\right\|_2 \leq \min\left\{\left(2\mu^{-1}\left\|\Delta_t\right\|_\infty \left\|M_t^0\right\|_1\right)^{1/2}, (1+\sqrt{2})\mu^{-1}\|\Delta_t\|_\infty\sqrt{r_t}\right\}. \tag{9}$$

Compared to (9), the bound (8) replaces the error term Δ_t by the smoothed version $\mathcal{W}_h\Delta_t$ at the cost of the bias $\delta_h M_t$.

To obtain an explicit expression of the error bound that may be used for bandwidth selection, we further investigate the properties of $\delta_h M_t$ and $\mathcal{W}_h\Delta_t$. Assumption 2 and 3 are standard conditions for temporal smoothness and the kernel function.

Assumption 2 *There exists a matrix function $M(t)$ supported on $[0,1]$, satisfying that*

1. *for any $j = 1, 2, \ldots, T$, $M_j^0 = M(j/T)$ and $\|M_j^0\|_1 \leq C_*$, $\max_{j,s,l}|[M_j^0]_{s,l}| \leq C_M$, for some constants C_* and C_M;*

2. *the first and second derivatives $\nabla M(t), \nabla^2 M(t)$ are continuous and bounded by some constants D_1 and D_2, respectively.*

Assumption 3 *The kernel function $K(x)$ is a symmetric probability density function on [-1, 1] and satisfies that*

$$\alpha(K) = \int x^2 K(x)\, dx < \infty, \quad R(K) = \int K(x)^2\, dx < \infty. \tag{10}$$

Define the $\psi-$norm $\|Z\|_{\psi(\eta)}$ for random variable Z with $0 < \eta < \infty$ as

$$\|Z\|_{\psi(\eta)} = \inf\left\{t > 0, \mathbb{E}\left[\exp\left(\frac{\|Z\|_\infty^\eta}{t^\eta}\right)\right] \leq 2\right\}$$

and $\|Z\|_{\psi(\infty)} = \inf\left\{t > 0, \mathbb{E}\mathbf{1}_{\{\|Z\|_\infty > t\}} = 0\right\}$. Then we impose the following assumption on the distributions of ξ_{ji} and X_{ji}.

Assumption 4 *The designs and noises at the same time point* $\{X_{ji}, \xi_{ji}\},\ i = 1, 2, \ldots, n_j$ *are mutually independent for each fixed* j *and there exists constants* $K_1, K_2 > 0,\ \alpha \geq 1,\ \beta \geq 2$ *that for any* $j = 1, 2 \ldots, T,\ i = 1, 2, \ldots, n_j$

1. ξ_{ji} *is sub-exponential distribution with* $\|\xi_{ji}\|_{\psi(\alpha)} \leq K_1$*;*

2. X_{ji} *is sub-gaussian distribution with* $\|X_{ji}\|_{\psi(\beta)} \leq K_2$*,*

3. *There exists* $\gamma \geq 1$ *such that* $1/\alpha + 1/\beta = 1/\gamma$.

The tail distributions of $\xi_{ji}X_{ji}$ and $\langle M, X_{ji}\rangle X_{ji}$ are both sub-exponential when X_{ji}, ξ_{ji} satisfy Assumption 4. Denote $\mathcal{D}_{\alpha,\beta,K_1,K_2}$ as the distribution set of $\{X_{ji}, \xi_{ji}\}_{ji}$ satisfying Assumption 4. We remark that $\mathcal{D}_{\alpha,\beta,K_1,K_2}$ is indeed general enough for applications. In the matrix completion problem of Example 1, X_{ji} are bounded with $\|X_{ji}\|_{\psi(\infty)} = 1$. Then the distributions of X_{ji}, ξ_{ji} lie in $\mathcal{D}_{1,\infty,K_1,1}$ as long as the noises ξ_{ji} follow sub-exponential distributions with $\alpha \geq 1$. In the compressed sensing problem of Example 2, from Theorem 4.4.5 in Vershynin (2018), we can set X_{ji} to be sub-guassian distributed with $\beta = 2$ and $K_2 \asymp (m_1 \vee m_2)^{1/2}\sigma_X$. Then the distributions of X_{ji}, ξ_{ji} lie in $\mathcal{D}_{2,2,K_1,\sigma_X}$ if noises ξ_{ji} follow a sub-guassian distributions with $\alpha = 2$.

Next we consider the temporal dependence between $\{X_{ji}, \xi_{ji}\}, i = 1, 2, \ldots, n_j$ and $\{X_{si}, \xi_{si}\}, i = 1, 2, \ldots, n_s$ that decays when the distance between time points j and s increases. To characterize the structure of such dependence, we impose the strongly dependent assumption that X_{ji} and ξ_{ji} are ϕ-mixing processes across j. Our definition of ϕ-mixing is the same as in Doukhan (2012), and the detail definition is given in Supplementary Material S.4. In the following subsections, we first derive the estimation error

bound in the ideal case that ξ_{ji} and X_{ji} are independent across time j, and then extend to the dependent case. These two settings are described in the following assumptions, respectively.

Assumption 5 *(Independent case) The designs and noises* $\{X_{ji}, \xi_{ji}\}$ *are mutually independent across* j *for each* i, $j = 1, 2, \ldots, T,\ i = 1, 2, \ldots, N_j$.

Assumption 6 *(ϕ-mixing processes) There exist two independent sequences of σ-fields $\mathcal{X}_j$ and $\mathcal{Y}_j, j \in \mathbb{N}^*$ such that*

$$\sigma((X_{j1}, X_{j2}, \ldots, X_{jn_j})^\top) \subseteq \mathcal{X}_j, \quad \sigma((\xi_{j1}, \xi_{j2}, \ldots, \xi_{jn_j})^\top) \subseteq \mathcal{Y}_j.$$

The σ-fields $\mathcal{X}_j$ and $\mathcal{Y}_j$, $j \in \mathbb{N}^*$ are both ϕ-mixing with coefficients $\phi_{\mathcal{X}}(k)$ and $\phi_{\mathcal{Y}}(k)$, $k \in \mathbb{N}$ satisfying that

$$\Phi_{\mathcal{X}} \triangleq \sum_{k=0}^{\infty} \sqrt{\phi_{\mathcal{X}}(k)} < \infty, \quad \Phi_{\mathcal{Y}} \triangleq \sum_{k=0}^{\infty} \sqrt{\phi_{\mathcal{Y}}(k)} < \infty.$$

3.2 Independent case

The theorem below states the upper bound of $\|\widetilde{M}_t^\lambda - M_t^0\|_2$ under the independent case when the distributions of X_{ji}, ξ_{ji} belong to $\mathcal{D}_{\alpha,\beta,K_1,K_2}$. For conciseness, we assume that the numbers of observations $n_j, j = 1, 2, \cdots, T$ have the same order denoted by n, i.e., $n_j \asymp n$ for $j = 1, 2, \cdots, T$. The conclusions for n_j of different orders can be derived similarly. In the following, the notations $a_n \ll b_n$, $a_n \gg b_n$ and $a_n \asymp b_n$ mean that the ratio a_n/b_n approaches to 0, infinity and a constant, respectively, and "$\lceil \cdot \rceil$" denotes the ceiling function.

Theorem 2 *Under Assumption 1–5, let* $n\lceil Th \rceil \gg (K_*/\sigma_*)^2 \log(m_1 + m_2)$, $h \to 0$ *and* $(m_1 m_2)^{1/2} \lceil T^2 h^2 \rceil T^{-1} \to \infty$ *as* $n, m_1, m_2, T \to \infty$, *when*

$$\lambda = 2C_1 \sigma_* \sqrt{\frac{\log(m_1 + m_2)}{n\lceil Th \rceil}},$$

where C_1 and σ_* are defined in Lemma S.4 of Supplementary Material, then with probability at least $1 - 3/(m_1 + m_2)$,

$$(m_1 m_2)^{-1/2} \left\| \widetilde{M}_t^{\lambda} - M_t^0 \right\|_2 \leq \frac{1}{2}\alpha(K) D_2 h^2 + \left(1 + \sqrt{2}\right) C_1 \sigma_* \left(\frac{r_t \log(m_1 + m_2)}{\mu^2 m_1 m_2 n \lceil Th \rceil} \right)^{1/2} + o(h^2), \tag{11}$$

where $\alpha(K)$ is defined in (10) *and D_2 in Assumption 2.*

We remark that $n\lceil Th \rceil$ is the number of samples used in a local window and the condition $n\lceil Th \rceil \gg (K_*/\sigma_*)^2 \log(m_1 + m_2)$ is the requirement for the effective sample size to guarantee the upper bound (11) tending to zero. The error bound (11) is a trade-off between the bias and variance which is controlled by the bandwidth h. When $n\mu^2 m_1 m_2/(T^4 \sigma_*^2 r_t \log(m_1 + m_2)) = o(1)$, we can choose the optimal bandwidth

$$h = C_h \left(\frac{\sigma_*^2 r_t \log(m_1 + m_2)}{\mu^2 m_1 m_2 n T} \right)^{1/5}, \tag{12}$$

where $C_h = \left[(2 + 2\sqrt{2}) C_1/(\alpha(K) D_2)\right]^{2/5}$ is a constant. When $n\mu^2 m_1 m_2/(T^4 \sigma_*^2 r_t \log(m_1 + m_2)) = \mathcal{O}(1)$, we choose $h = 0$ which is degenerate to the classical situation with the convergence rate $n^{-1/2}$.

We focus on a large enough value for T in the subsequent discussion, which allows us to omit the ceiling operator $\lceil \cdot \rceil$. For situations where T is small, we can simply choose $h = 0$, which yields the same results as static methods. With an appropriately selected h, we have the following corollary.

Corollary 1 *Under assumptions of Theorem 2, when*

$$nT \gg \max\{K_*^{5/2} \sigma_*^{-3} (\mu^2 m_1 m_2)^{1/4} \log(m_1 + m_2),\ \sigma_*^2 (\mu^2 m_1 m_2)^{-1} \log(m_1 + m_2)\},$$

with probability at least $1 - 3/(m_1 + m_2)$,

$$(m_1 m_2)^{-1/2} \left\| \widetilde{M}_t^{\lambda} - M_t^0 \right\|_2 \leq C_2 \left(\frac{\sigma_*^2 r_t \log(m_1 + m_2)}{\mu^2 m_1 m_2 n T} \right)^{2/5}, \tag{13}$$

where $C_2 = 1/2\left[(2+2\sqrt{2})C_1\right]^{4/5}\left[\alpha(K)D_2\right]^{1/5}$.

The bound requirement for nT is composed of two terms: the first is to make $nTh \gg (K_*/\sigma_*)^2 \log(m_1+m_2)$ hold when h takes the form (12), and the second is to guarantee the optimal bandwidth (12) tends to 0 as $n, T \to \infty$. In (13), the upper bound is proportional to $(nT)^{-2/5}$ where nT is the total sample size. This indicates that our dynamic method makes efficient use of information from adjacent time points to improve the estimation quality. The number of observations in a single time point n and the number of time points T complement each other to reduce the upper bound (13). When the samples are extremely sparse at each time point, it is infeasible for the classical static method to reconstruct the eigenvalues and eigenvectors of M_t^0. However, the proposed method can still control the estimation error $\|\widetilde{M}_t^\lambda - M_t^0\|_2$ to a desirable level, as long as there are sufficient time points. We also mention that the two-step smoothing method performs poorly, because the first-step estimate $\widehat{M}_t^\lambda$ and the reconstruction of eigen-subspace at each time point are far from satisfaction, which cannot be fixed by the second smoothing step regardless how dense the time points are. We emphasize that Theorem 2 and Corollary 1 can be applied directly to any dynamic trace regression problems as long as X_{ji} and ξ_{ji} lie in $\mathcal{D}_{\alpha,\beta,K_1,K_2}$. Taking dynamic matrix completion for example, we have the following corollary. The corresponding result for dynamic compressed sensing can be derived similarly, which is presented in Corollary S.3 of the Supplement.

Corollary 2 *Under Assumption 1–5, when X_{ji} are i.i.d. uniformly distributed on $\mathcal{E}$, ξ_{ji} are independently follow sub-exponential mean-zero distributions, $nTh \gg (m_1 \wedge m_2)\log^{1+2/\alpha}(m_1+m_2)$, and $(m_1m_2)^{1/2}Th^2 \to \infty$ as $h \to 0, n, m_1, m_2, T \to \infty$, then with probability at least*

$1 - 3/(m_1 + m_2)$,

$$(m_1 m_2)^{-1/2} \left\| \widetilde{M}_t^{\lambda} - M_t^0 \right\|_2$$
$$\leq \frac{1}{2}\alpha(K) D_2 h^2 + (1+\sqrt{2}) C_1 (C_M \vee \sigma_\xi) \left(\frac{r_t (m_1 \vee m_2) \log(m_1 + m_2)}{nTh} \right)^{1/2} + o(h^2).$$

When $nT \gg (m_1 \vee m_2) \log^{1+5/(2\alpha)}(m_1 + m_2)$, *let*

$$h = C_h \left(\frac{(C_M \vee \sigma_\xi)^2 r_t (m_1 \vee m_2) \log(m_1 + m_2)}{nT} \right)^{1/5}, \tag{14}$$

then

$$(m_1 m_2)^{-1/2} \left\| \widetilde{M}_t^{\lambda} - M_t^0 \right\|_2 \leq C_2 (C_M \vee \sigma_\xi)^{4/5} \left(\frac{r_t (m_1 \vee m_2) \log(m_1 + m_2)}{nT} \right)^{2/5}, \tag{15}$$

where C_1, C_h, C_2 *are the same constants as above.*

Recall that the classical static result for the error bound of matrix completion in Koltchinskii et al. (2011a) is

$$(m_1 m_2)^{-1/2} \left\| \widehat{M}_t^{\lambda} - M_t^0 \right\|_2 \leq (1+\sqrt{2}) C_1 (C_M \vee \sigma_\xi) \left(\frac{r_t (m_1 \vee m_2) \log(m_1 + m_2)}{n} \right)^{1/2}.$$

The dynamic method provides a sharper upper bound than the classic one. In other words, when the sample size in each single time point is sparse, our method can borrow information from adjacent time points to improve the estimation, as long as the number of time points T is sufficiently large.

3.3 Dependent case

In parallel to the independent case, we derive the bounds of $\|\widetilde{M}_t^{\lambda} - M_t^0\|_2$ when ξ_{ji}, X_{ji} are ϕ-mixing processes, as stated in Theorem 3 below.

Theorem 3 *With Assumption 1–4 and 6, when* $Th \gg \log(m_1+m_2)$, $nTh \gg (K_*/\sigma_*)^2 \log^3(m_1+m_2)$, $h \to 0$ *and* $(m_1m_2)^{1/2}Th^2 \to \infty$ *as* $n, m_1, m_2, T \to \infty$, *with probability at least* $1 - 3/(m_1+m_2)$,

$$(m_1m_2)^{-1/2} \left\| \widetilde{M}_t^{\lambda} - M_t^0 \right\|_2 \leq \frac{1}{2}\alpha(K)D_2h^2 + (1+\sqrt{2})\mathcal{C}_1(\Phi_{\mathcal{X}} \vee \Phi_{\mathcal{Y}})\sigma_* \left(\frac{r_t \log(m_1+m_2)}{\mu^2 m_1 m_2 nTh} \right)^{1/2} + o(h^2) \tag{16}$$

where $\mathcal{C}_1 > 0$ *is a constant defined in Lemma S.5 of Supplementary Material.*

We use $\Phi_{\mathcal{X}}$ and $\Phi_{\mathcal{Y}}$ to measure the temporal dependence of design matrices X_{ji} and noises ξ_{ji}, respectively, where larger $\Phi_{\mathcal{X}}$ and $\Phi_{\mathcal{Y}}$ mean stronger dependence. When $\Phi_{\mathcal{X}} = \Phi_{\mathcal{Y}} = 1$, it indicates that X_{ji}, ξ_{ji} are independent across j and the bound (16) is the same as (11) in Theorem 2 in the independent case. When $nT \gg (\mu^2 m_1 m_2)^{-1}\sigma_*^2 \log(m_1+m_2)$, if we choose

$$h = \mathcal{C}_h \left(\frac{\sigma_*^2(\Phi_{\mathcal{X}} \vee \Phi_{\mathcal{Y}})^2 r_t \log(m_1+m_2)}{\mu^2 m_1 m_2 nT} \right)^{1/5}, \tag{17}$$

where $\mathcal{C}_h = \left[(2+\sqrt{2})\mathcal{C}_1/(\alpha(K)D_2)\right]^{2/5}$, then we have the following corollary.

Corollary 3 *Under assumptions of Theorem 3, when*

$$n^{-1}T^4 \gg \sigma_*^{-2}\mu^2 m_1 m_2 \log^4(m_1+m_2)$$

and

$$nT \gg \max\{K_*^{5/2}\sigma_*^{-3}(\mu^2 m_1 m_2)^{1/4}\log(m_1+m_2),\ \sigma_*^2(\mu^2 m_1 m_2)^{-1}\log(m_1+m_2)\},$$

with probability at least $1 - 3/(m_1+m_2)$,

$$(m_1m_2)^{-1/2} \left\| \widetilde{M}_t^{\lambda} - M_t^0 \right\|_2 \leq \mathcal{C}_2 \left(\frac{\sigma_*^2(\Phi_{\mathcal{X}} \vee \Phi_{\mathcal{Y}})^2 r_t \log(m_1+m_2)}{\mu^2 m_1 m_2 nT} \right)^{2/5}, \tag{18}$$

where $\mathcal{C}_2 = 1/2\left[(2+\sqrt{2})\mathcal{C}_1\right]^{4/5}\left[\alpha(K)D_2\right]^{1/5}$.

When X_{ji} and/or ξ_{ji} have strong long-term dependence across j, $\Phi_{\mathcal{X}} \vee \Phi_{\mathcal{Y}}$ becomes large and inflates the bound (18), which means that the convergence rate of $\widetilde{M}_t^{\lambda}$ is usually slower than that in independent case. Note that the bound for the single-stage estimates at each t using static method can also be derived from Theorem 2 by using a degenerated kernel $\delta_0(x)$, given by

$$(m_1 m_2)^{-1/2} \left\| \widehat{M}_t^{\lambda} - M_t^0 \right\|_2 \leq (1+\sqrt{2}) C_1 \sigma_* \left(\frac{r_t \log(m_1+m_2)}{\mu^2 m_1 m_2 n} \right)^{1/2}.$$

Then when

$$\Phi_{\mathcal{X}} \vee \Phi_{\mathcal{Y}} \ll T^{1/2} \left(\frac{\sigma^{*2} r_t \log(m_1+m_2)}{\mu^2 m_1 m_2 n} \right)^{1/8}, \tag{19}$$

our dynamic method yields a sharper bound than the static one though strong dependence across time exists. With the expression of h in (17), the condition (19) is equal to $Th \gg (\Phi_{\mathcal{X}} \vee \Phi_{\mathcal{Y}})^2$. Note that Th corresponds to the effective time points used in estimation at each t. This implies that as the number of effective time points increases, the negative effect caused by temporal correlation would be offset to some extent. For space economy, the applications of Theorem 3 and Corollary 3 to dynamic matrix completion and dynamic compressed sensing are presented as Corollary S.2 and Corollary S.4 in Supplementary Material.

3.4 Empirical error and complexity analysis

Recall that the sequence generated by the iteration of Algorithm 1 at time t are denoted by $\{M_t^{(k)}\}_k, k \in \mathbb{N}^*$, where $M_t^{(k)}$ is the output matrix after k-th iteration and $M_t^{(0)}$ is the initial input matrix. In practice, we use the output $M_t^{(k)}$ to approximate M_t^0 for a certain k. Thus, we derive the upper bound of the empirical error $\left\| M_t^{(k)} - M_t^0 \right\|_2$ in the following theorem, which takes both the algorithmic error between $M_t^{(k)}$ and $\widetilde{M}_t^{\lambda}$ and the estimation error between $\widetilde{M}_t^{\lambda}$ and M_t^0 into account.

Theorem 4 *Under assumptions of Theorem 1, for any $k > 1$ and $\lambda \geq 2\|\mathcal{W}_h \Delta_t\|_\infty$, we have*

$$\left\| M_t^{(k)} - M_t^0 \right\|_2 \leq \delta_h M_t + \min \left\{ \left((\delta_h M_t)^2 + \frac{2\lambda}{\mu} \|M_t^0\|_1 + \frac{2L_f \gamma_t^2}{\mu (k+1)^2} \right)^{1/2}, \right.$$
$$\left. \frac{1+\sqrt{2}}{2} \frac{\lambda}{\mu} \sqrt{r_t} + \left(\frac{2L_f \gamma_t^2}{\mu (k+1)^2} \right)^{1/2} \right\}$$

where $\gamma_t = \left\| M_t^{(0)} - \widetilde{M}_t^\lambda \right\|_2$ and $L_f = 2 \left\| \sum_{j=1}^T \omega_h(j-t) \cdot \frac{1}{n_j} \sum_{i=1}^{n_t} \left(\|X_{ji}\|_2 X_{ji} \right) \right\|_2$ is the Lipschitz constant of gradient function $\nabla f_t(M)$ that is finite under Assumption 4.

The empirical error bound in Theorem 4 has an additional term $2L_f \gamma_t^2 / \mu (k+1)^2$ compared to the estimation error bound (7) in Theorem 1, which is the so-called algorithmic error. The optimal choice is to stop iterating when the algorithmic error attains the same order as the estimation error. Let k_t be the minimal number of iteration steps needed to attain the desired order. From the expression of the algorithmic error, k_t depends on the distance between the initial value $M_t^{(0)}$ and the target $\widetilde{M}_t^\lambda$, denoted by γ_t. Since less iterations are required for a smaller initial distance, the naive strategy to initialize $M_t^{(0)}$ randomly at each time t would lead a large γ_t. Noting that $\widetilde{M}_t^\lambda$ is close to $\widetilde{M}_{t-1}^\lambda$, we thus make use of the neighboring information in the initialization to reduce computation cost. Hence we propose to use the output matrix at time $(t-1)$ as the initial value for estimating $\widetilde{M}_t^\lambda$ at time t, i.e. $M_t^{(0)} = M_{t-1}^{(k_{t-1})}$. Take the dependent case in Section 3.3 for instance. Recall that the optimal estimation error bound is derived in Corollary 3 with appropriate choice of the bandwidth h. Then desired empirical error bound for Theorem 4 can be derived by

$$(m_1 m_2)^{-1/2} \left\| M_t^{(k_t)} - M_t^0 \right\|_2 \leq 2\mathcal{C}_2 \left(\frac{\sigma_*^2 (\Phi_\mathcal{X} \vee \Phi_\mathcal{Y})^2 r_t \log(m_1 + m_2)}{\mu^2 m_1 m_2 n T} \right)^{2/5}. \tag{20}$$

To quantify the improvement of our initialization strategy, in Corollary 4, we compute the total number of iteration steps to attain the bound (20) at each t, and compare with that of

the random initialization. The result for independent case can be derived similarly, which is presented in Corollary S.5 of Supplementary Material for space economy.

Corollary 4 *Under the assumptions of Theorem 3, when the total number of iterations of random initialization K_0 satisfies*

$$K_0 \geq \mathcal{C}_2^{-1} \left(\frac{\sigma_*^2 (\Phi_{\mathcal{X}} \vee \Phi_{\mathcal{Y}})^2 r_t \log(m_1 + m_2)}{\mu^2 m_1 m_2} \right)^{-2/5} \left(\frac{2L_f}{\mu m_1 m_2} \right)^{1/2} \gamma_1 n^{2/5} T^{7/5},$$

or total number of iterations of the proposed initial strategy K_1 satisfies

$$\begin{aligned} K_1 \geq & \mathcal{C}_2^{-1} \left(\frac{\sigma_*^2 (\Phi_{\mathcal{X}} \vee \Phi_{\mathcal{Y}})^2 r_t \log(m_1 + m_2)}{\mu^2 m_1 m_2} \right)^{-2/5} \left(\frac{2L_f}{\mu m_1 m_2} \right)^{1/2} (\gamma_1 + D_1)(nT)^{2/5} \\ & + 3\sqrt{2} \left(\frac{L_f}{\mu} \right)^{1/2} T. \end{aligned}$$

the empirical error is lower than the optimal upper bound (20) *for each $t = 1, \ldots, T$.*

From Corollary 4, once the total number of iterations of random initialization attains the level $\mathcal{O}(n^{2/5}T^{7/5})$ and that of Algorithm 1 attains $\mathcal{O}(n^{2/5}T^{2/5}+T)$, the error is bounded by (20). Then the ratio of computational cost of our proposed algorithm compared to the random initialization method is $\mathcal{O}(T^{-1} \vee n^{-2/5}T^{-2/5})$. We finally mention that the classical static method need $\mathcal{O}(n^{1/2})$ iterations to attain the optimal rate $\mathcal{O}(n^{-1/2})$ for each t, and hence the single-stage method (at all T times) need $\mathcal{O}(n^{1/2}T)$ iterations in total. Therefore the ratio of computational cost of Algorithm 1 compared to the single-stage method is $\mathcal{O}(T^{-3/5} \vee n^{-1/2})$, which leads to the conclusion that our algorithm is computationally more efficient than the random initial method and the classical static methods.

4 Simulation Studies

We take the dynamic matrix completion as the instance to verify the usefulness of our method. To demonstrate the advantages, we compare the proposed estimates with three

benchmark methods: (i) the classical static estimator $\widehat{M}_t^{\lambda}$ defined as in (4), abbreviated as Static; (ii) the two-step smoothing estimator $\widehat{M}_t^{\text{t-s}}$ given in (5), abbreviated as TwoStep; (iii) the low-rank tensor completion estimator, abbreviated as Tensor,

$$\widehat{N}^{\lambda} = \underset{N \in \mathbb{R}^{T \times m_1 \times m_2}}{\operatorname{argmin}} \|\mathcal{P}_{\Omega}(N) - Y\|_2^2 + \lambda \sum_{i=1}^{3} \|N_{(i)}\|_1,$$

where $\mathcal{P}_{\Omega}$ is the projection to the observation subspace Ω, Y is the observed data tensor with element

$$[Y]_{j,s,l} = \begin{cases} \dfrac{1}{|\mathcal{A}_{j,s,l}|} \displaystyle\sum_{i \in \mathcal{A}_{j,s,l}} Y_{ji}, & \text{if } \mathcal{A}_{j,s,l} \neq \emptyset \\ 0, & \text{otherwise} \end{cases}$$

for $\mathcal{A}_{j,s,l} = \left\{ i \mid X_{ji} = e_s(m_1) e_l^{\top}(m_2), 1 \leq i \leq n_j \right\}$, and $N_{(i)}$ is the mode-i unfolding of tensor N which is a matrix that arranges the mode-i fibers to be columns. We use matrix form FISTA algorithm in Toh & Yun (2010) to approximate $\widehat{M}_t^{\lambda}$, and $\widehat{M}_t^{\text{t-s}}$ has a closed form solution. In Tensor, we use AMD-TR(E) algorithm in Gandy et al. (2011) to numerically approximate $\widehat{N}^{\lambda}$.

We set $m_1 = 500, m_2 = 300, r_t = 10$ and the matrix function $M(t)$ in Assumption 2 is constructed by $M(t) = U(t)D(t)V^T(t)$ with

$$\begin{aligned} U(t) &= \cos\left(\frac{t\pi}{2}\right) U_0 + \sin\left(\frac{t\pi}{2}\right) U_1, \\ D(t) &= 10 \left(\operatorname{diag}\left\{10^2, 9^2, \ldots, 1\right\} + t \operatorname{diag}\left\{10, 9, \ldots, 1\right\} \right), \\ V(t) &= \cos\left(\frac{t\pi}{2}\right) V_0 + \sin\left(\frac{t\pi}{2}\right) V_1, \end{aligned}$$

where $U_0, U_1 \in \mathbb{R}^{m_1 \times r_t}$, $V_0, V_1 \in \mathbb{R}^{m_2 \times r_t}$ are composed of standard orthogonal basis. Here we consider the situation that each n_t is the same, i.e., $n_t = n, t = 1, 2, \cdots, T$. Denote the rate of observation samples as $\rho = n/(m_1 m_2)$ and the density of time points as $\tau = 1/T$. We use a 5-fold cross validation to select λ in the numerical experiments. Please refer to Section S.9.4 of the Supplement for more details.

For the independent case, we sample $X_{ji},\ i = 1, 2, \ldots, n$ from $\mathcal{E}$ defined in (3) independently with uniform probabilities, and let $\xi_{ji}, j = 1, 2, \ldots, T,\ i = 1, 2, \ldots, n$ i.i.d follow the normal distribution $N(0, \sigma_\xi^2)$, where $T = 100$ and $\sigma_\xi = 1$. We use the Epanechnikov kernel and set the bandwidth h by the optimal selection form (14) in which we approximate $C_M \vee \sigma_\xi$ by mean of the top 10% largest Y_{ji} and D_2 by $\sum_{j=2}^{T} |1/n_j \sum_{i=1}^{n_j} Y_{ji} - 1/n_{j-1} \sum_{i=1}^{n_{j-1}} Y_{j-1i}|$. We set $\rho = 0.2$ for our proposed dynamic low-rank (DLR) and Tensor methods. In Static and TwoStep, $\widehat{M}_t^{\lambda}$ and $\widehat{M}_t^{\text{t-s}}$ fail to recover the matrix structure under the same setting $\rho = 0.2$, so we set $\rho = 0.8$ for feasibility. Figure 1 illustrates the mean square errors

$$\text{MSE}_t = (m_1 m_2)^{-1} \left\| \widehat{M}_t - M_t^0 \right\|_2^2, \tag{21}$$

for $t = 1, 2, \cdots, T$. For the proposed DLR and TwoStep methods, the MSE_t floats up at the boundaries due to slight edge effect. The average MSE, i.e., $T^{-1} \sum_{t=1}^{T} \text{MSE}_t$ for the DLR estimate is 0.085 and for Tensor is 0.44 when $\rho = 0.2$. This implies that the tensor recovery is not effective at least when the temporal structure is smooth. For Static and TwoStep, when $\rho = 0.8$, the average MSE is 0.16 and 0.13 respectively. That is, these two methods with four times data still perform worse than the proposed DLR method. This verifies the theoretical finding that when the sample size n at a single time point is small, the classical trace regression fails to estimate M_t^0, while our dynamic method can borrow adjacent information to improve the estimation quality.

Now we study the empirical influence of ρ and τ on the estimation results. One can see from (15) that given the underlying matrix, the bound of MSE is in proportion to $(\rho/\tau)^{-4/5}$. We set eight different values for (ρ, τ) which can be divided into three groups, i.e. $\rho/\tau = 5, 10, 20$. The MSE_t across t under these settings are shown in the left panel of Figure 2, which indicates that given the matrix dimension, the empirical MSE depends on ρ/τ. We further plot the logarithm of average MSE across time versus the logarithm of

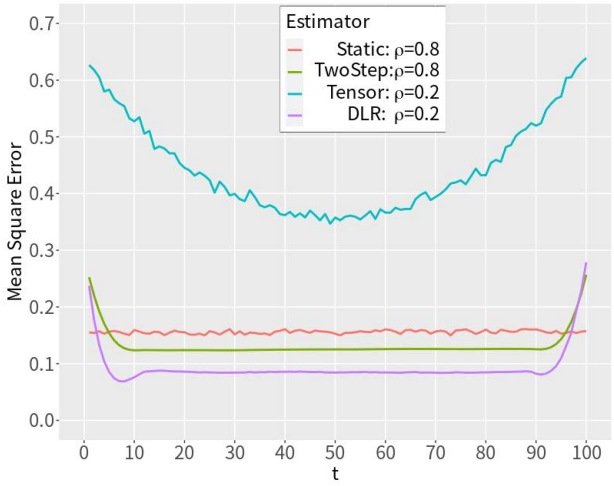


Figure 1: The comparisons of MSE_t for different methods. In DLR and Tensor, $\rho = 0.2$, and in Static and TwoStep, $\rho = 0.8$.

ρ/τ in the right panel of Figure 2, which reveals the linear relationship of the slope -4/5 between them. This further validates the conclusion (15).

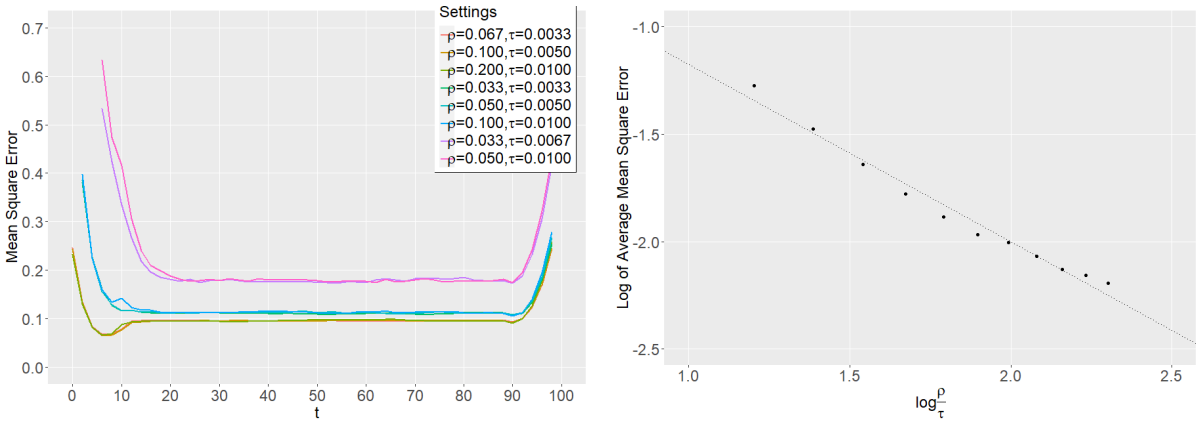


Figure 2: The left panel corresponds to MSE_t of DLR estimates under different settings of ρ and τ. The curves are clustered into three classes corresponding to $\rho/\tau = 5, 10, 20$. The right panel shows the logarithm of average MSE versus the logarithm of ρ/τ showing a linear trend with the slope -4/5.

We also investigate the influence of the dependent noise on the proposed estimator. The noises ξ_{ji} are generated by

$$
\begin{aligned}
&E_0 = U_0, \quad U_0 \sim \mathrm{WN}(0, \sigma_\xi^2 \mathbf{1}_{m_1} \mathbf{1}_{m_2}^T), \\
&E_t = \beta E_{t-1} + \sqrt{1-\beta^2} U_t, \quad U_t \sim \mathrm{WN}(0, \sigma_\xi^2 \mathbf{1}_{m_1} \mathbf{1}_{m_2}^T), \\
&\xi_{ji} = \langle E_j, X_{ji} \rangle, \quad j = 1, 2, \ldots, T, \ i = 1, 2, \ldots, n_t,
\end{aligned} \tag{22}
$$

where $\mathrm{WN}(0, \sigma_\xi^2 \mathbf{1}_{m_1} \mathbf{1}_{m_2}^T)$ generates the $m_1 \times m_2$ random matrix whose elements are i.i.d. normal with mean zero and variance σ_ξ^2. The larger β is, the stronger dependence exists among $\{\xi_{ji}\}_j$. The other parameters are set are the same as those in the independent case. Then $\Phi_{\mathcal{X}} = 1$ and $\Phi_{\mathcal{Y}}$ can be calculated from (22) which is larger than 1 when $\beta > 0$. We apply the proposed method to recover M_t^0 for $1 \le t \le T$ under the combinations of $\sigma_\xi = 1, 2$ and $\beta = 0, 0.3, 0.6, 0.9$. As illustrated in the left panel of Figure 3, the MSE increases as σ_ξ^2 and/or β enlarge(s). Note that according to (S.26) in Supplement, the MSE is proportional to $\Phi_{\mathcal{Y}}^{8/5}$ given M_t, n, T. We plot the average MSE versus $\Phi_{\mathcal{Y}}^{8/5}$ under different σ_ξ^2 in the right panel of Figure 3 which verifies the conclusion.

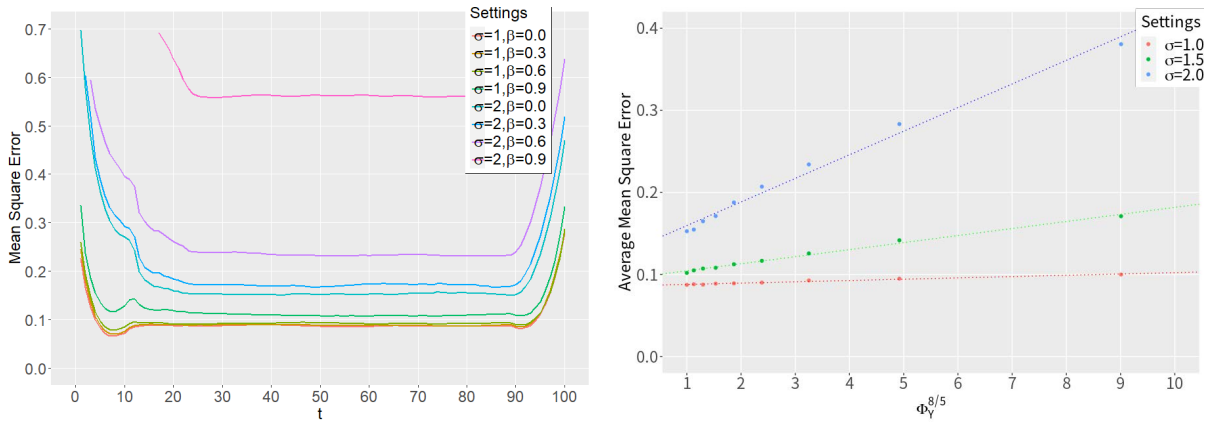


Figure 3: The left panel illustrates the MSE_t across time with different σ_ξ and β and the right panel shows that the average MSE versus $\Phi_{\mathcal{Y}}^{8/5}$ under different settings when $\Phi_{\mathcal{X}} = 1$.

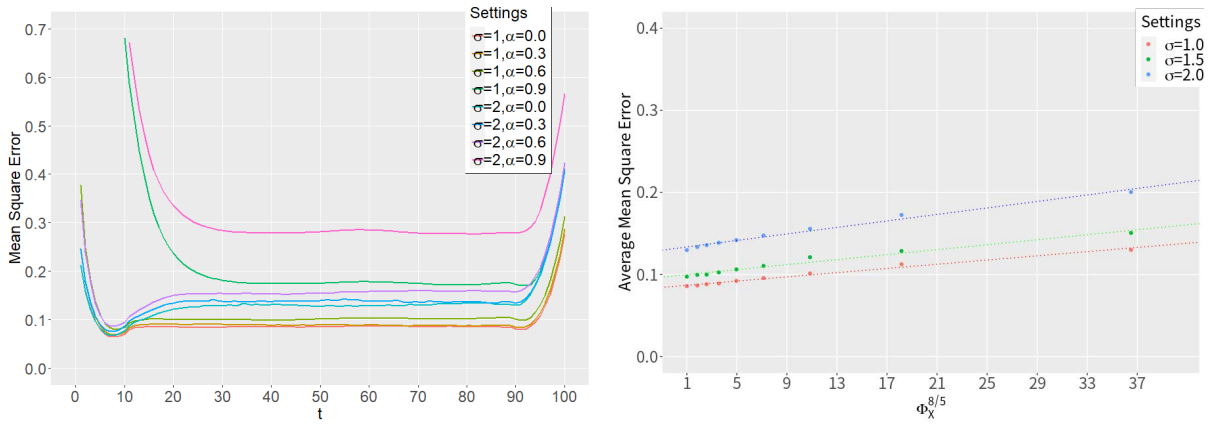


Figure 4: The left panel illustrates the MSE_t across time with different σ_ξ and α and the right panel shows that the average MSE versus $\Phi_{\mathcal{X}}^{8/5}$ under different settings when $\Phi_{\mathcal{Y}} = 1$.

Finally we study the influence of the dependence of design matrices on the estimation. We sample $X_{1i}, i = 1, \ldots, n$ from $\mathcal{E}$ with uniform probabilities independently and generate $X_{ji}, j \geq 2, i = 1, 2, \ldots, n$ by preserving αn elements randomly in $X_{j-1,i}, i = 1, 2, \ldots, n$ and choosing $(1-\alpha)n$ new elements i.i.d. from $\mathcal{E}$. Then the dependence of design matrices across time becomes stronger as α increases. We set the remaining parameters the same as those in the independent case and have $\Phi_{\mathcal{Y}} = 1$, $\Phi_{\mathcal{X}} \geq 1$, where the equality holds if and only if $\alpha = 0$. We plot the MSE_t across time under different $\sigma_\xi = 1, 2$ and $\alpha = 0, 0.3, 0.6, 0.9$ and verify the linear relationship between the average MSE and $\Phi_{\mathcal{X}}^{8/5}$ in Figure 4. We remark that the slopes of linear relationships in the right panels of Figure 3 and 4 are different. This is because the bound (S.26) in Supplement is obtained by the technique that $C_M \Phi_{\mathcal{X}} + \sigma_\xi \Phi_{\mathcal{Y}} \leq (C_M \vee \sigma_\xi)(\Phi_{\mathcal{X}} \vee \Phi_{\mathcal{Y}})$. Hence the curves of the average MSE versus $\Phi_{\mathcal{Y}}^{8/5}$ become sharper as σ_ξ increases, as shown in the right panel of Figure 3, while the slopes of the average MSE versus $\Phi_{\mathcal{X}}^{8/5}$ under different σ_ξ in the right panel of Figure 4 are similar.

5 Real Data Examples

In this section, we apply the proposed method to two real data examples for the dynamic matrix completion and compressed sensing problems, respectively.

The first example is the Netflix Prize Dataset (Netflix 2006), in which users ratings for movies received by Netflix were collected from October 1998 to December 2005 and can be downloaded at https://www.kaggle.com/datasets/netflix-inc/netflix-prize-data. The ratings are integer-valued from 1 to 5. We preprocess the dataset to remove those movies which are watched less than 25000 times.

For user selection, we consider two different filters. Filter 1 is to choose users whose rating times are more than 700 because users who watch a larger number of movies tend to offer more reliable ratings and make better comparisons across different movies due to their broader exposure. There remain 3036 users and 1034 movies with total 2337997 ratings. We utilize a highly efficient data merging approach that requires minimal computational resources. Specifically, we divide the observations into $T = 100$ chronological time intervals. Within each interval, observations are aggregated as they occur at distinct time points. Then we randomly choose 4/5 of them in each interval as training data and set the rest as test data. At each time point t, there exists an underlying matrix $M_t \in \mathbb{R}^{m_1 \times m_2}$ to be recovered, where the rows represent users and the columns represent movies. The observed sparse matrix $Y_t \in \mathbb{R}^{m_1 \times m_2}$ has elements at position (j, k), representing the rating given by user j to movie k at time point t, or zero if no rating is available. Thus we can convert it to the trace regression model. The average sample size at each time point is about 17803 and thus the rate of observation samples in each time point is $\rho \approx 0.0057$. We use again the Epanechnikov kernel $K(x)$ and select the bandwidth h in the same way as in the simulations. Filter 2 is to select users whose rating times are more than 30 because those

users can represent the target groups of users who often using Netflix to watch movies. Then we conduct a random selection of 3000 individuals from the pool of eligible users for ease of computation.

We use 5-fold cross validation to choose tuning parameter λ and evaluate the performance of $\widetilde{M}_t^{\lambda}$ by calculating

$$\mathrm{MSE}_t^* = \frac{1}{n_t^{\mathrm{test}}} \sum_{i=1}^{n_t^{\mathrm{test}}} \left(\langle X_{t,i}^{\mathrm{test}}, \widetilde{M}_t^{\lambda} \rangle - Y_{t,i}^{\mathrm{test}} \right)^2, \tag{23}$$

using the test data and compare the proposed method with those three benchmarks mentioned in Section 4. Figure 5 shows that our dynamic method is more accurate than those benchmarks across the time domain. And the average MSE^*, i.e. $T^{-1}\sum_{t=1}^{T}\mathrm{MSE}_t^*$, of our dynamic method and benchmarks Static, TwoStep and Tensor methods are compared in Table 1.

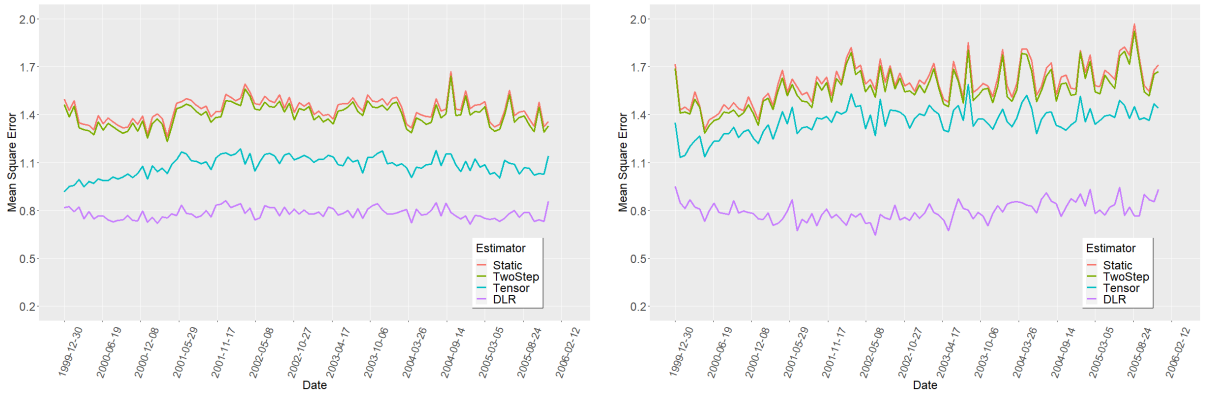


Figure 5: The MSE_t^* (23) at different time points for the proposed method and three benchmarks for the Netflix dataset. The left plot is the result for Filter 1 and the right plot is the result for Filter 2.

Another example is the compression and recovery of videos. We use the lion video from Davis 2017 dataset (Pont-Tuset et al. 2017) and treat each frame as a matrix at each time point. The dataset at https://davischallenge.org/davis2017/code.html#unsupervised

Table 1: The average MSE* for Netflix dataset.

Setting	DLR	Static	TwoStep	Tensor
Filter 1 MSE*	0.781	1.431	1.397	1.082
Filter 2 MSE*	0.796	1.601	1.565	1.360

is publicly available. For compression, we first separate the matrices M_t^0 into the sparse part S_t and the low-rank part L_t using robust principal component analysis (Candès et al. 2011). The design matrices X_{ti} are the convolution kernel matrices with random centers (a_{ti}, b_{ti}), i.e., $[X_{ti}]_{a_{ti},b_{ti}} = 4, [X_{ti}]_{a_{ti}\pm1,b_{ti}} = [X_{ti}]_{a_{ti},b_{ti}\pm1} = 2, [X_{ti}]_{a_{ti}\pm1,b_{ti}\pm1} = 1$. It is straightforward to verify that the distributions of $\{X_{ti}\}_{t,i}$ satisfy Assumption 4. Then we compress the low-rank part to obtain corresponding output $Y_{ti} = \langle X_{ti}, L_t \rangle$. For recovery, we use the dynamic trace regression to reconstruct the low-rank part, denoted as $\widetilde{L}_t^\lambda$. Then we recover each frame by $\widetilde{M}_t^\lambda = S_t + \widetilde{L}_t^\lambda$, while the sparse part S_t is retained. We emphasize that we only need store the centers (a_{ti}, b_{ti}), the output Y_{ti} and the nonzero part of S_t to recover the matrices M_t^0, which is considerably space-saving compared to storing the whole video. In the lion video, each frame has 480×854 pixels. We generate Y_{ti} with the rate of observation samples $\rho = 0.146$ and the file volume is reduced by 70% after compression. Here we use Static and TwoStep mentioned in Section 4 to reconstruct $\widehat{L}_t^\lambda, \widehat{L}_t^{\text{t-s}}$ and recover $\widehat{M}_t^\lambda = S_t + \widehat{L}_t^\lambda$, $\widehat{M}^{\text{t-s}} = S_t + \widehat{L}_t^{\text{t-s}}$ as benchmarks. Figure 6 shows the original frames M_t^0 and the estimates $\widetilde{M}_t^\lambda, \widehat{M}_t^\lambda, \widehat{M}^{\text{t-s}}$ at different time points. The background and the smooth motions of all three lions are recovered well by the proposed DLR method, which can be seen visually better than two benchmark methods. Recall the definition (23), the average $\text{MSE}^* = T^{-1}\sum_{t=1}^T \text{MSE}_t^*$ is 2.9×10^{-3} for our dynamic method while is 7.6×10^{-2} and 4.7×10^{-2} for the two benchmarks Static and TwoStep repsectively.



Figure 6: The top five pictures are the 5th, 25th, 45th, 65th and 85th original frames in the lions video, the second rows are the corresponding estimates of the proposed DLR method and the third and fourth rows are the benchmarks Static and TwoStep, respectively.

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6 Proof of Theorem 1

We need the following lemma for proving Theorem 1.

Lemma 1 *Let $\Sigma \in \mathbb{S}_+^{m\times m}$ be a positive definite matrix with the smallest eigenvalue $\mu > 0$. Then the following inequality*

$$\alpha^\top \Sigma(\alpha+\beta) \geq \mu\left(\|\alpha\|_2^2 - \kappa_\Sigma \|\alpha\|_2\|\beta\|_2\right) \tag{24}$$

holds for any $\alpha, \beta \in \mathbb{R}^m$

Proof of Lemma 1: This follows immediately from Assumption 1, since $\alpha^\top\Sigma\alpha \geq \mu\|\alpha\|_2^2$ and $\alpha^\top\Sigma\beta \geq -\kappa_\Sigma\mu\|\alpha\|_2\|\beta\|_2$. □

Proof of Theorem 1: For brevity, we denote

$$\omega_j = \omega_h(j-t), \quad \mathcal{C}_j(M,N) = \frac{1}{n_j}\sum_{i=1}^{n_j}\mathbb{E}\langle M, X_{ji}\rangle\langle N, X_{ji}\rangle$$

for all $M, N \in \mathbb{R}_{m_1\times m_2}$. With the definition of $\widetilde{M}_t^\lambda$,

$$\begin{aligned}
&\sum_{j=1}^T \omega_j \mathcal{C}_j(\widetilde{M}_t^\lambda, \widetilde{M}_t^\lambda) - 2\sum_{j=1}^T \omega_j \left\langle \frac{1}{n_j}\sum_{i=1}^{n_j} Y_{ji}X_{ji}, \widetilde{M}_t^\lambda \right\rangle + \lambda\|\widetilde{M}_t^\lambda\|_1 \\
\leq &\sum_{j=1}^T \omega_j \mathcal{C}_j(M,M) - 2\sum_{j=1}^T \omega_j \left\langle \frac{1}{n_j}\sum_{i=1}^{n_j} Y_{ji}X_{ji}, M \right\rangle + \lambda\|M\|_1
\end{aligned}$$

holds for any $M \in \mathbb{R}^{m_1\times m_2}$. With the identity

$$\left\langle \frac{1}{n_j}\sum_{i=1}^{n_j} Y_{ji}X_{ji}, M \right\rangle = \langle \Delta_j, M\rangle + \mathcal{C}_j(M_j^0, M),$$

we have

$$\begin{aligned}
&\sum_{j=1}^T \omega_j \mathcal{C}_j(\widetilde{M}_t^\lambda, \widetilde{M}_t^\lambda) - 2\sum_{j=1}^T \omega_j \mathcal{C}_j(M_j^0, \widetilde{M}_t^\lambda) \\
\leq &\sum_{j=1}^T \omega_j \mathcal{C}_j(M,M) - 2\sum_{j=1}^T \omega_j \mathcal{C}_j(M_j^0, M) + 2\sum_{j=1}^T \omega_j \langle \Delta_j, \widetilde{M}_t^\lambda - M\rangle + \lambda(\|M\|_1 - \|\widetilde{M}_t^\lambda\|_1).
\end{aligned}$$

Using $\langle \cdot, * \rangle \leq \| \cdot \|_\infty \| * \|_1$, one can obtain

$$\begin{aligned}
&\sum_{j=1}^{T} \omega_j \mathcal{C}_j(\widetilde{M}_t^\lambda - M_j^0, \widetilde{M}_t^\lambda - M_j^0) \\
\leq &\sum_{j=1}^{T} \omega_j \mathcal{C}_j(M - M_j^0, M - M_j^0) + 2\| \sum_{j=1}^{T} \omega_j \Delta_j \|_\infty \|\widetilde{M}_t^\lambda - M\|_1 + \lambda(\|M\|_1 - \|\widetilde{M}_t^\lambda\|_1).
\end{aligned}$$

Under the assumption that $2\| \sum_{j=1}^{T} \omega_j \Delta_j \|_\infty \leq \lambda$, we have

$$\sum_{j=1}^{T} \omega_j \mathcal{C}_j(\widetilde{M}_t^\lambda - M_j^0, \widetilde{M}_t^\lambda - M_j^0) \leq \sum_{j=1}^{T} \omega_j \mathcal{C}_j(M - M_j^0, M - M_j^0) + 2\lambda \|M\|_1.$$

Set $M = M_t^0$, then

$$\sum_{j=1}^{T} \omega_j \mathcal{C}_j(\widetilde{M}_t^\lambda - M_j^0, \widetilde{M}_t^\lambda - M_j^0) \leq \sum_{j=1}^{T} \omega_j \mathcal{C}_j(M_t^0 - M_j^0, M_t^0 - M_j^0) + 2\lambda \|M_t^0\|_1.$$

With the identity

$$\begin{aligned}
&\mathcal{C}_j(\widetilde{M}_t^\lambda - M_j^0, \widetilde{M}_t^\lambda - M_j^0) \\
=&\mathcal{C}_j(\widetilde{M}_t^\lambda - M_t^0, \widetilde{M}_t^\lambda - M_t^0) + \mathcal{C}_j(M_t^0 - M_j^0, M_t^0 - M_j^0) + 2\mathcal{C}_j(\widetilde{M}_t^\lambda - M_t^0, M_t^0 - M_j^0),
\end{aligned}$$

we have

$$\sum_{j=1}^{T} \omega_j \mathcal{C}_j(\widetilde{M}_t^\lambda - M_t^0, \widetilde{M}_t^\lambda - M_t^0) + 2\sum_{j=1}^{T} \omega_j \mathcal{C}_j(\widetilde{M}_t^\lambda - M_t^0, M_t^0 - M_j^0) \leq 2\lambda \|M_t^0\|_1. \tag{25}$$

If $\left\| \widetilde{M}_t^\lambda - M_t^0 \right\|_2 < 2 \left\| M_t^0 - \sum_{j=1}^{T} \omega_j M_j^0 \right\|_2$, we can maintain the bound

$$\left\| \widetilde{M}_t^\lambda - M_t^0 \right\|_2 < 2\delta_h M_t. \tag{26}$$

If $\left\| \widetilde{M}_t^\lambda - M_t^0 \right\|_2 \geq 2 \left\| M_t^0 - \sum_{j=1}^{T} \omega_j M_j^0 \right\|_2$, with Assumption 1, we know that

$$\mathcal{C}_j(M, N) = \operatorname{vec}(M)^\top \Sigma \operatorname{vec}(N). \tag{27}$$

So (25) can be expressed as

$$\begin{aligned}
&\left(\operatorname{vec}(\widehat{M}_t^\lambda) - \operatorname{vec}(M_t^0) \right)^\top \Sigma \left[\left(\operatorname{vec}(\widehat{M}_t^\lambda) - \operatorname{vec}(M_t^0) \right) + \left(2\operatorname{vec}(M_t^0) - 2\sum_{j=1}^{T} \omega_j \operatorname{vec}(M_j^0) \right) \right] \\
\leq &2\lambda \|M_t^0\|_1.
\end{aligned}$$

From Lemma 1,

$$\begin{aligned}&\left(\operatorname{vec}(\widehat{M}_t^\lambda)-\operatorname{vec}(M_t^0)\right)^\top\Sigma\left[\left(\operatorname{vec}(\widehat{M}_t^\lambda)-\operatorname{vec}(M_t^0)\right)+\left(2\operatorname{vec}(M_t^0)-2\sum_{j=1}^T\omega_j\operatorname{vec}(M_j^0)\right)\right]\\ \geq&\mu\left\|\widetilde{M}_t^\lambda-M_t^0\right\|_2^2-2\mu\kappa_\Sigma\left\|\widetilde{M}_t^\lambda-M_t^0\right\|_2\left\|M_t^0-\sum_{j=1}^T\omega_jM_j^0\right\|_2.\end{aligned}\tag{28}$$

Thus we have

$$\mu\left\|\widetilde{M}_t^\lambda-M_t^0\right\|_2^2-2\mu\kappa_\Sigma\left\|\widetilde{M}_t^\lambda-M_t^0\right\|_2\left\|M_t^0-\sum_{j=1}^T\omega_jM_j^0\right\|_2-2\lambda\|M_t^0\|_1\leq 0.$$

By solving this inequality, we have

$$\left\|\widetilde{M}_t^\lambda-M_t^0\right\|_2\leq\frac{4\mu\kappa_\Sigma\delta_hM_t+\sqrt{\mu^2(\delta_hM_t)^2+8\mu\lambda\|M_t^0\|_1}}{2\mu}\lesssim\delta_hM_t+\sqrt{\frac{2\lambda\|M_t^0\|_1}{\mu}}.\tag{29}$$

Combining (26) and (29),

$$\left\|\widetilde{M}_t^\lambda-M_t^0\right\|_2\lesssim\delta_hM_t+\max\left\{\delta_hM_t,\sqrt{\frac{2\lambda\|M_t^0\|_1}{\mu}}\right\}\leq\delta_hM_t+\left((\delta_hM_t)^2+2\mu^{-1}\lambda\|M_t^0\|_1\right)^{1/2}.\tag{30}$$

Because $\widetilde{M}_t^\lambda$ is the minimizer of (6), there exists a sub-gradient matrix B satisfying that for all $M\in\mathbb{R}^{m_1\times m_2}$,

$$\left\langle B,\widetilde{M}_t^\lambda-M\right\rangle\leq 0,$$

which equals to that there exists $\widehat{V}_t\in\partial\|\widetilde{M}_t^\lambda\|_1$ satisfying that

$$2\sum_{j=1}^T\omega_j\left(\mathcal{C}_j(\widetilde{M}_t^\lambda,\widetilde{M}_t^\lambda-M)-\left\langle\frac{1}{n_j}\sum_{i=1}^{n_j}Y_{ji}X_{ji},\widetilde{M}_t^\lambda-M\right\rangle\right)+\lambda\langle\widehat{V}_t,\widetilde{M}_t^\lambda-M\rangle\leq 0.$$

It is equivalent to

$$\begin{aligned}&2\sum_{j=1}^T\omega_j\mathcal{C}_j(\widetilde{M}_t^\lambda-M_j^0,\widetilde{M}_t^\lambda-M)+\lambda\langle\widehat{V}_t-V,\widetilde{M}_t^\lambda-M\rangle\\ \leq&-\lambda\langle V,\widetilde{M}_t^\lambda-M\rangle+2\sum_{j=1}^T\omega_j\langle\Delta_j,\widetilde{M}_t^\lambda-M\rangle,\end{aligned}$$

where $V \in \partial\|M\|_1$.

Denote that $M = \sum_{j=1}^{r} \sigma_j u_j v_j^\top$ with support S_1, S_2. V have representation from Watson (1992):

$$V = \sum_{j=1}^{r} u_j v_j^\top + P_{S_1^\perp} W P_{S_2^\perp},\ \|W\|_\infty \le 1.$$

We choose a W subject to

$$\langle P_{S_1^\perp} W P_{S_2^\perp}, \widetilde{M}_t^\lambda - M\rangle = \langle W, P_{S_1^\perp}\widehat{M}_t^\lambda P_{S_2^\perp}\rangle = \|P_{S_1^\perp}\widehat{M}_t^\lambda P_{S_2^\perp}\|_1.$$

With the monotonicity of sub-gradients, we have $\langle \widehat{V}_t - V, \widetilde{M}_t^\lambda - M\rangle \ge 0$. So

$$\begin{aligned} &2\sum_{j=1}^{T} \omega_j \mathcal{C}_j(\widetilde{M}_t^\lambda - M_j^0, \widetilde{M}_t^\lambda - M) + \lambda\|P_{S_1^\perp}\widetilde{M}_t^\lambda P_{S_2^\perp}\|_1 \\ \le& -\lambda\langle\sum_{j=1}^{r} u_j v_j^\top, \widetilde{M}_t^\lambda - M\rangle + 2\sum_{j=1}^{T}\omega_j\langle\Delta_j, \widetilde{M}_t^\lambda - M\rangle. \end{aligned}$$

Using the identity

$$\begin{aligned} &2\sum_{j=1}^{T} \omega_j \mathcal{C}_j(\widetilde{M}_t^\lambda - M_j^0, \widetilde{M}_t^\lambda - M) \\ =&\sum_{j=1}^{T}\omega_j\left(\mathcal{C}_j(\widetilde{M}_t^\lambda - M_j^0, \widetilde{M}_t^\lambda - M_j^0) + \mathcal{C}_j(\widetilde{M}_t^\lambda - M, \widetilde{M}_t^\lambda - M) - \mathcal{C}_j(M - M_j^0, M - M_j^0)\right) \end{aligned}$$

and the inequality

$$-\langle\sum_{j=1}^{r} u_j v_j^\top, \widetilde{M}_t^\lambda - M\rangle = -\langle\sum_{j=1}^{r} u_j v_j^\top, P_{S_1}(\widetilde{M}_t^\lambda - M)P_{S_2}\rangle \le \|P_{S_1}(\widetilde{M}_t^\lambda - M)P_{S_2}\|_1,$$

we have

$$\begin{aligned} &\sum_{j=1}^{T} \omega_j \mathcal{C}_j(\widetilde{M}_t^\lambda - M_j^0, \widetilde{M}_t^\lambda - M_j^0) + \sum_{j=1}^{T} \omega_j \mathcal{C}_j(\widetilde{M}_t^\lambda - M, \widetilde{M}_t^\lambda - M) + \lambda\|P_{S_1^\perp}\widetilde{M}_t^\lambda P_{S_2^\perp}\|_1 \\ \le&\sum_{j=1}^{T} \omega_j \mathcal{C}_j(M - M_j^0, M - M_j^0) + \lambda\|P_{S_1}(\widetilde{M}_t^\lambda - M)P_{S_2}\|_1 + 2\sum_{j=1}^{T}\omega_j\langle\Delta_j, \widetilde{M}_t^\lambda - M\rangle. \end{aligned} \tag{31}$$

With identity

$$\begin{aligned} \langle\Delta_j, \widetilde{M}_t^\lambda - M\rangle &= \langle P_M(\Delta_j), \widehat{M}_t^\lambda - M\rangle + \langle P_{S_1^\perp}\Delta_j P_{S_2^\perp}, \widehat{M}_t^\lambda - M\rangle \\ &= \langle P_M(\Delta_j), \widehat{M}_t^\lambda - M\rangle + \langle P_{S_1^\perp}\Delta_j P_{S_2^\perp}, \widehat{M}_t^\lambda\rangle, \end{aligned}$$

we have that

$$
\begin{aligned}
&\sum_{j=1}^{T}\omega_j\langle\Delta_j,\widetilde{M}_t^{\lambda}-M\rangle\\
=&\left\langle P_M\left(\sum_{j=1}^{T}\omega_j\Delta_j\right),\widehat{M}_t^{\lambda}-M\right\rangle+\left\langle P_{S_1^{\perp}}\left(\sum_{j=1}^{T}\omega_j\Delta_j\right)P_{S_2^{\perp}},\widehat{M}_t^{\lambda}\right\rangle\\
\le&\left\|P_M\left(\sum_{j=1}^{T}\omega_j\Delta_j\right)\right\|_2\left\|\widetilde{M}_t^{\lambda}-M\right\|_2+\left\|P_{S_1^{\perp}}\left(\sum_{j=1}^{T}\omega_j\Delta_j\right)P_{S_2^{\perp}}\right\|_{\infty}\left\|P_{S_1^{\perp}}\widetilde{M}_t^{\lambda}P_{S_2^{\perp}}\right\|_1.
\end{aligned}
$$

With

$$
\begin{aligned}
\left\|P_M\left(\sum_{j=1}^{T}\omega_j\Delta_j\right)\right\|_2&\le\sqrt{\operatorname{rank}\left[P_A\left(\sum_{j=1}^{T}\omega_j\Delta_j\right)\right]}\left\|P_M\left(\sum_{j=1}^{T}\omega_j\Delta_j\right)\right\|_{\infty}\\
&\le\sqrt{2\operatorname{rank}(M)}\|\sum_{j=1}^{T}\omega_j\Delta_j\|_{\infty}
\end{aligned}
$$

and

$$
\left\|P_{S_1^{\perp}}\left(\sum_{j=1}^{T}\omega_j\Delta_j\right)P_{S_2^{\perp}}\right\|_{\infty}\le\|\sum_{j=1}^{T}\omega_j\Delta_j\|_{\infty},
$$

we have

$$
\sum_{j=1}^{T}\omega_j\langle\Delta_j,\widetilde{M}_t^{\lambda}-M\rangle\le\sqrt{2\operatorname{rank}(M)}\|\sum_{j=1}^{T}\omega_j\Delta_j\|_{\infty}\|\widetilde{M}_t^{\lambda}-M\|_2+\|\sum_{j=1}^{T}\omega_j\Delta_j\|_{\infty}\|P_{S_1^{\perp}}\widetilde{M}_t^{\lambda}P_{S_2^{\perp}}\|_1. \tag{32}
$$

Also we have

$$
\begin{aligned}
\|P_{S_1}(\widetilde{M}_t^{\lambda}-M)P_{S_2}\|_1&\le\sqrt{\operatorname{rank}(M)}\|P_{S_1}(\widetilde{M}_t^{\lambda}-M)P_{S_2}\|_2\\
&\le\sqrt{\operatorname{rank}(M)}\|\widetilde{M}_t^{\lambda}-M\|_2.
\end{aligned} \tag{33}
$$

With (31), (32) and (33),

$$
\begin{aligned}
&\sum_{j=1}^{T}\omega_j\left[\mathcal{C}_j(\widetilde{M}_t^{\lambda}-M_j^0,\widetilde{M}_t^{\lambda}-M_j^0)+\mathcal{C}_j(\widetilde{M}_t^{\lambda}-M,\widetilde{M}_t^{\lambda}-M)\right]\\
+&\left(\lambda-2\|\sum_{j=1}^{T}\omega_j\Delta_j\|_{\infty}\right)\|P_{S_1^{\perp}}\widetilde{M}_t^{\lambda}P_{S_2^{\perp}}\|_1\\
\le&\sum_{j=1}^{T}\omega_j\mathcal{C}_j(M-M_j^0,M-M_j^0)+\sqrt{\operatorname{rank}(M)}\left(\lambda+2\sqrt{2}\|\sum_{j=1}^{T}\omega_j\Delta_j\|_{\infty}\right)\|\widetilde{M}_t^{\lambda}-M\|_2.
\end{aligned}
$$

With $\lambda \geq 2\|\sum_{j=1}^{T}\omega_j\Delta_j\|_\infty$,

$$\begin{aligned}
&\sum_{j=1}^{T}\omega_j\mathcal{C}_j(\widetilde{M}_t^\lambda - M_j^0, \widetilde{M}_t^\lambda - M_j^0) + \sum_{j=1}^{T}\omega_j\mathcal{C}_j(\widetilde{M}_t^\lambda - M, \widetilde{M}_t^\lambda - M)\\
\leq &\sum_{j=1}^{T}\omega_j\mathcal{C}_j(M - M_j^0, M - M_j^0) + \lambda(1+\sqrt{2})\sqrt{\text{rank}(M)}\|\widetilde{M}_t^\lambda - M\|_2.
\end{aligned}$$

Set $M = M_t^0$, so

$$\begin{aligned}
&\sum_{j=1}^{T}\omega_j\mathcal{C}_j(\widetilde{M}_t^\lambda - M_j^0, \widetilde{M}_t^\lambda - M_j^0) + \sum_{j=1}^{T}\omega_j\mathcal{C}_j(\widetilde{M}_t^\lambda - M_t^0, \widetilde{M}_t^\lambda - M_t^0)\\
\leq &\sum_{j=1}^{T}\omega_j\mathcal{C}_j(M_t^0 - M_j^0, M_t^0 - M_j^0) + \lambda(1+\sqrt{2})\sqrt{\text{rank}(M)}\|\widetilde{M}_t^\lambda - M_t^0\|_2.
\end{aligned}$$

With the identity

$$\mathcal{C}_j(\widetilde{M}_t^\lambda - M_j^0, \widetilde{M}_t^\lambda - M_j^0) = \mathcal{C}_j(\widetilde{M}_t^\lambda - M_t^0) + 2\mathcal{C}_j(\widetilde{M}_t^\lambda - M_t^0, M_t^0 - M_j^0) + \mathcal{C}_j(M_t^0 - M_j^0, M_t^0 - M_j^0),$$

we have

$$\begin{aligned}
&\sum_{j=1}^{T}\omega_j\left[\mathcal{C}_j(\widetilde{M}_t^\lambda - M_t^0, \widetilde{M}_t^\lambda - M_t^0) + \mathcal{C}_j(\widetilde{M}_t^\lambda - M_t^0, M_t^0 - M_j^0)\right]\\
\leq &\frac{1+\sqrt{2}}{2}\lambda\sqrt{\text{rank}(M_t^0)}\|\widetilde{M}_t^\lambda - M_t^0\|_2.
\end{aligned}$$

If $\|\widetilde{M}_t^\lambda - M_t^0\|_2 < \left\|M_t^0 - \sum_{j=1}^{T}\omega_j M_j^0\right\|_2$, we maintain the bound

$$\|\widetilde{M}_t^\lambda - M_t^0\|_2 < \delta_h M_t. \tag{34}$$

If $\|\widetilde{M}_t^\lambda - M_t^0\|_2 \geq \left\|M_t^0 - \sum_{j=1}^{T}\omega_j M_j^0\right\|_2$, using (27) and Lemma 1, we know that

$$\begin{aligned}
&\sum_{j=1}^{T}\omega_j\left[\mathcal{C}_j(\widetilde{M}_t^\lambda - M_t^0, \widetilde{M}_t^\lambda - M_t^0) + \mathcal{C}_j(\widetilde{M}_t^\lambda - M_t^0, M_t^0 - M_j^0)\right]\\
= &\left(\text{vec}(\widehat{M}_t^\lambda) - \text{vec}(M_t^0)\right)^\top \Sigma\left(\left(\text{vec}(\widehat{M}_t^\lambda) - \text{vec}(M_t^0)\right) + \left(M_t^0 - \sum_{j=1}^{T}\omega_j M_j^0\right)\right) \qquad (35)\\
\geq &\mu\left\|\widetilde{M}_t^\lambda - M_t^0\right\|_2^2 - \mu\kappa_\Sigma\left\|\widetilde{M}_t^\lambda - M_t^0\right\|_2\left\|M_t^0 - \sum_{j=1}^{T}\omega_j M_j^0\right\|_2.
\end{aligned}$$

So we have

$$\mu \left\| \widetilde{M}_t^{\lambda} - M_t^0 \right\|_2^2 - \mu\kappa_{\Sigma} \left\| \widetilde{M}_t^{\lambda} - M_t^0 \right\|_2 \left\| M_t^0 - \sum_{j=1}^{T} \omega_j M_j^0 \right\|_2 \leq \frac{1+\sqrt{2}}{2} \lambda \sqrt{\operatorname{rank}(M_t^0)} \left\| \widetilde{M}_t^{\lambda} - M_t^0 \right\|_2 . \tag{36}$$

Combining (34) and (36), we have

$$\left\| \widetilde{M}_t^{\lambda} - M_t^0 \right\|_2 \lesssim \delta_h M_t + \frac{1+\sqrt{2}}{2} \frac{\lambda}{\mu} \sqrt{r_t}, \tag{37}$$

where $r_t = \operatorname{rank}(M_t^0)$. With (30) and (37), the proof is completed.

□

7 Proofs of Theorem 2 and Corollary 1, 2

Proposition 1 *Under Assumption 2 and 3, if* $h \to 0$ *and* $(m_1 m_2)^{1/2} T h^2 \to \infty$ *as* $m_1, m_2, T \to \infty$*, then*

$$\delta_h M_t \leq \frac{1}{2} \alpha(K) (m_1 m_2)^{1/2} D_2 h^2 + o\left((m_1 m_2)^{1/2} h^2 \right).$$

Proof of Proposition 1: Note that $T^{-1} \ll (m_1 m_2)^{-1/2} h^2$. The conclusion follows from the classical nonparametric results (of fixed design) that

$$\begin{aligned}
\left\| M_t^0 - \sum_{j=1}^{T} \omega_j M_j^0 \right\|_2 &= \left\| \sum_{j=1}^{T} \frac{1}{Th} K\left(\frac{j-t}{Th} \right) \left\{ M\left(\frac{j}{T} \right) - M\left(\frac{t}{T} \right) \right\} \right\|_2 \\
&= \left\| \int \frac{1}{h} K\left(\frac{u - t/T}{h} \right) \left\{ M(u) - M\left(\frac{t}{T} \right) \right\} du \right\|_2 + O\left(T^{-1} \right) \\
&= \left\| \int K(z) \left\{ M\left(zh + \frac{t}{T} \right) - M\left(\frac{t}{T} \right) \right\} dz \right\|_2 + O\left(T^{-1} \right) \\
&= \frac{1}{2} \alpha(K) \left\| \nabla^2 M(\frac{t}{T}) \right\|_2 h^2 + o\left((m_1 m_2)^{1/2} h^2 \right) \\
&\leq \frac{1}{2} \alpha(K) (m_1 m_2)^{1/2} D_2 h^2 + o\left((m_1 m_2)^{1/2} h^2 \right).
\end{aligned}$$

□

Now we prove two lemmas which are needed to finish the proofs Lemma 4 and Theorem 2. For brevity, define that $\|\cdot\| \triangleq \|\cdot\|_\infty$. We extend proposition 2 in Koltchinskii (2011b) to obtain the following Lemma 2.

Lemma 2 *Let $Z_1, Z_2, \ldots, Z_n \in (\mathcal{B}(\mathbb{R}^{m_1\times m_2}), \mathbb{R}^{m_1\times m_2}, \mu^{\otimes m_1 m_2})$ be independent, mean-zero random variables satisfying that*

$$\|Z_i\|_{L^2} \le \sigma, \quad \|Z_i\|_{\psi(\alpha)} \le K, \quad \|\cdot\|_{\psi(\alpha)} := \inf\left\{t > 0, \mathbb{E}\left[\exp\left(\frac{\|\cdot\|^\alpha}{t^\alpha}\right)\right] \le 2\right\}.$$

Let $\alpha > 1$ and $a = (a_1, a_2, \ldots, a_n)^\top \in \mathbb{R}^n$. There exists a constant $C > 0$ such that, for all $t > 0$, with probability at least $1 - e^{-t}$

$$\left\|\sum_{i=1}^n a_i Z_i\right\| \le C \max\left(\sqrt{\|a\|_2^2 \sigma^2 (t + \log(2m))}, \|a\| K \left(\log\frac{K}{\sigma}\right)^{1/\alpha} (t + \log(2m))\right). \quad (38)$$

Proof of Lemma 2: We follow the proof of proposition 2 in Koltchinskii (2011b) here.

Let

$$S_n := \sum_{i=1}^n a_i \widetilde{Z}_i, \quad \widetilde{Z} = \begin{pmatrix} 0 & Z \\ Z^\top & 0 \end{pmatrix} \in \mathbb{R}^{m\times m}.$$

We know that $\|S_n\| < s$ if and only if $-sI_m \prec S_n \prec sI_m$. Therefore,

$$P(\|S_n\| \ge s) = P(S_n \not\preceq sI_m) + P(S_n \not\succeq -sI_m)$$

and

$$P(S_n \not\preceq sI_m) = P\left(e^{\lambda S_n} \not\preceq e^{\lambda s I_m}\right) \le P\left(\operatorname{Tr}(e^{\lambda S_n}) \ge e^{\lambda s}\right) \le e^{-\lambda s}\mathbb{E}\operatorname{Tr}(e^{\lambda S_n}).$$

To bound $\mathbb{E}\operatorname{Tr}(e^{\lambda S_n})$, we use independence and Golden-Thompson inequality

$$\begin{aligned}
\mathbb{E}\operatorname{Tr}(e^{\lambda S_n}) &= \mathbb{E}\operatorname{Tr}\left(e^{\lambda S_{n-1} + \lambda a_n Z_n}\right) \\
&\le \mathbb{E}\operatorname{Tr}\left(e^{\lambda S_{n_1}} e^{\lambda a_n Z_n}\right) \\
&= \operatorname{Tr}\left(\mathbb{E}e^{\lambda S_{n-1}} \mathbb{E}e^{\lambda a_n Z_n}\right) \\
&\le \mathbb{E}\operatorname{Tr}\left(e^{\lambda S_{n-1}}\right) \left\|\mathbb{E}e^{\lambda a_n Z_n}\right\|.
\end{aligned}$$

By induction, we have

$$\mathbb{E}\,\mathrm{Tr}(e^{\lambda S_n}) \le m \prod_{i=1}^{n} \left\| \mathbb{E} e^{\lambda a_i Z_i} \right\|.$$

It remains to bound the norm $\|\mathbb{E}e^{\lambda a_i Z_i}\|$. Using Taylor expansion and $\mathbb{E}(Z_i) = 0$, we have

$$\begin{aligned}
\mathbb{E}e^{\lambda a_i Z_i} &= I_m + \mathbb{E}\lambda^2 a_i^2 Z_i^2 \left[\frac{1}{2!} + \frac{\lambda a_i Z_i}{3!} + \frac{\lambda^2 a_i^2 Z_i^2}{4!} + \dots \right] \\
&\le I_m + \lambda^2 a_i^2 \mathbb{E} Z_i^2 \left[\frac{1}{2!} + \frac{\lambda a_i \|Z_i\|}{3!} + \frac{\lambda^2 a_i^2 \|Z_i\|^2}{4!} + \dots \right] \\
&= I_m + \lambda^2 a_i^2 \mathbb{E} Z_i^2 \left[\frac{e^{\lambda a_i \|Z_i\|} - 1 - \lambda a_i \|Z_i\|}{\lambda^2 a_i^2 \|Z_i\|^2} \right].
\end{aligned}$$

Therefore, for all $\tau > 0$ we have

$$\begin{aligned}
&\left\| \mathbb{E} e^{\lambda a_i Z_i} \right\| \\
&\le 1 + \lambda^2 a_i^2 \left\| \mathbb{E} Z_i^2 \left[\frac{e^{\lambda a_i \|Z_i\|} - 1 - \lambda a_i \|Z_i\|}{\lambda^2 a_i^2 \|Z_i\|^2} \right] \right\| \\
&\le 1 + \lambda^2 \left\| \mathbb{E} a_i^2 Z_i^2 \right\| \left[\frac{e^{\lambda\tau} - 1 - \lambda\tau}{\lambda^2 \tau^2} \right] + \lambda^2 \mathbb{E} \|a_i Z_i\|^2 \left[\frac{e^{\lambda a_i \|Z_i\|} - 1 - \lambda a_i \|Z_i\|}{\lambda^2 a_i^2 \|Z_i\|^2} \right] I(\|a_i Z_i\| \ge \tau).
\end{aligned}$$

Let

$$\tau = C_2 \|a_i\|_\infty K (\log \frac{K}{\sigma})^{1/\alpha}, \quad \lambda\tau \le 1,$$

we have

$$\left\| \mathbb{E} e^{\lambda a_i Z_i} \right\| \le 1 + C_1 \lambda^2 a_i^2 \sigma^2 \le \exp\{C_1 \lambda^2 a_i^2 \sigma^2\}.$$

So when

$$P(\|S_n\| \ge s) \le 2m \exp\{-\lambda s + C_1 \lambda^2 \|a\|_2^2 \sigma^2\}$$

and λ is chosen to be

$$\min \left\{ \frac{1}{C_2 \|a\|_\infty K \left(\log K/\sigma\right)^{1/\alpha}}, \frac{s}{2C_1 \|a\|_2^2 \sigma^2} \right\},$$

which can directly deduce (38).

□

Lemma 3 *If X, Y are random variables with $\|X\|_{\psi(\alpha)} \leq K_1$, $\|Y\|_{\psi(\beta)} \leq K_2$, and there exists $\gamma \geq 1$ that $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{1}{\gamma} \leq 1$, then we have*

$$\|XY\|_{\psi(\gamma)} \leq K_1 K_2. \tag{39}$$

Proof of Lemma 3: Using Young's inequality we know that for all $\theta \in (0,1)$

$$\frac{(\|X\|^\gamma)^{\alpha/\gamma}}{t^{\alpha\theta}\alpha/\gamma} + \frac{(\|Y\|^\gamma)^{\beta/\gamma}}{t^{\beta(1-\theta)}\beta/\gamma} \geq \frac{(\|XY\|)^\gamma}{t^\gamma}.$$

Thus we have

$$\begin{aligned}\mathbb{E}\left[\exp\left(\frac{\|XY\|^\gamma}{t^\gamma}\right)\right] &\leq \mathbb{E}\left[\exp\left(\frac{(\|X\|^\gamma)^{\alpha/\gamma}}{t^{\alpha\theta}\alpha/\gamma} + \frac{(\|Y\|^\gamma)^{\beta/\gamma}}{t^{\beta(1-\theta)}\beta/\gamma}\right)\right] \\ &\leq \frac{\gamma}{\alpha}\mathbb{E}\left[\exp\left(\frac{(\|X\|^\alpha}{t^{\alpha\theta}}\right)\right] + \frac{\gamma}{\beta}\mathbb{E}\left[\exp\left(\frac{(\|Y\|^\beta}{t^{\beta(1-\theta)}}\right)\right] \quad \leq 2,\end{aligned}$$

where we choose t, θ by $t^\theta = K_1, t^{1-\theta} = K_2$.

So we have

$$\mathbb{E}\left[\exp\left(\frac{\|XY\|^\gamma}{(K_1K_2)^\gamma}\right)\right] \leq 2.$$

□

We construct the following theorem to bound $\|\sum_{j=1}^T \omega_h(j-t)\Delta_j\|$ and Lemma 4 is an immediate result of the following theorem. For brevity, we denote $\omega_j = \omega_h(j-t)$.

Theorem 5 *Under Assumption 4-5, for all $t > 0$, with probability at least $1 - 3e^{-t}$,*

$$\begin{aligned}\left\|\sum_{j=1}^T \omega_j \Delta_j\right\| &\leq C_1 \max\left(\sigma\sqrt{\frac{(t+\log(2m))}{N}}, \frac{K\left(\log\frac{K}{\sigma}\right)^{1/\gamma}(t+\log(2m))}{W}\right) \\ &\quad + C_2\mu_X \max\left(\sigma_\xi\sqrt{\frac{t}{N}}, \frac{K_1(\log\frac{K_1}{\sigma_\xi})^{1/\alpha}t}{W}\right) \\ &\quad + C_3 \max\left(\varsigma\sqrt{\frac{(t+\log(2m))}{N}}, \frac{\mathcal{K}\left(\log\frac{\mathcal{K}}{\varsigma}\right)^{2/\beta}(t+\log(2m))}{W}\right),\end{aligned}$$

where

$$N = \frac{1}{\sum_{j=1}^T \omega_j^2/n_j}, \quad W = \frac{1}{\max_j \omega_j/n_j}$$

and constants with respect to ω_j, n_j such that

$$K = \max \|\xi_{ji}(X_{ji} - \mathbb{E}X_{ji})\|_{\psi(\gamma)} \lesssim K_1 K_2,$$

$$\sigma = \max \|\xi_{ji}(X_{ji} - \mathbb{E}X_{ji})\|_{L^2} \lesssim K_1 K_2 2^{1/\gamma},$$

$$\mu_X = \max \|\mathbb{E}X_{ji}\| \lesssim K_2,$$

$$\sigma_\xi = \max \|\xi_{ji}\|_{L^2} \lesssim K_1 2^{1/\alpha},$$

$$\mathcal{K} = \max \|\langle M_j^0, X_{ji}\rangle X_{ji} - \mathbb{E}(\langle M_j^0, X_{ji}\rangle X_{ji})\|_{\psi(\beta/2)} \lesssim C_* K_2^2,$$

$$\varsigma = \max \|\langle M_j^0, X_{ji}\rangle X_{ji} - \mathbb{E}(\langle M_j^0, X_{ji}\rangle X_{ji})\|_{L^2} \lesssim C_* K_2^2 2^{2/\beta}.$$

Proof of Theorem 5: Because that

$$\begin{aligned}
\left\|\sum_{j=1}^{T} \omega_j \Delta_j\right\| &= \left\|\sum_{j=1}^{T}\sum_{i=1}^{n_j} \frac{1}{n_j}\omega_j \left(Y_{ji}X_{ji} - EY_{ji}X_{ji}\right)\right\| \\
&\leq \left\|\sum_{j=1}^{T}\sum_{i=1}^{n_j} \frac{1}{n_j}\omega_j \xi_{ji} X_{ij}\right\| \\
&+ \left\|\sum_{j=1}^{T}\sum_{i=1}^{n_j} \frac{1}{n_j}\omega_j \left\{\langle M_0(j), X_{ji}\rangle X_{ji} - E\left(\langle M_0(j), X_{ji}\rangle X_{ji}\right)\right\}\right\| \\
&\leq \left\|\sum_{j=1}^{T}\sum_{i=1}^{n_j} \frac{1}{n_j}\omega_j \xi_{ji}(X_{ij} - EX_{ij})\right\| + \left\|\sum_{j=1}^{T}\sum_{i=1}^{n_j} \frac{1}{n_j}\omega_j \xi_{ji}\mathbb{E}X_{ij}\right\| \\
&+ \left\|\sum_{j=1}^{T}\sum_{i=1}^{n_j} \frac{1}{n_j}\omega_j \left\{\langle M_0(j), X_{ji}\rangle X_{ji} - E\left(\langle M_0(j), X_{ji}\rangle X_{ji}\right)\right\}\right\| \\
&:= \mathcal{D}_1 + \mathcal{D}_2 + \mathcal{D}_3.
\end{aligned} \tag{40}$$

a. For $\mathcal{D}_1$.

We denote that $Z_{ji} := \xi_{ji}(X_{ji} - \mathbb{E}X_{ji})$. Using Lemma 3, Z_{ji} are independent, mean-zero random variables satisfying that

$$\|Z_{ji}\|_{\psi(\gamma)} \leq K, \quad \|Z_{ji}\|_{L^2} \leq \sigma(2)^{1/\gamma}.$$

Set $a_{ji} = \omega_j/n_j$. Using Lemma 2, we have that for all $t > 0$, with probability at least

$1-e^{-t}$

$$\mathcal{D}_1 = \left\| \sum_{j=1}^{T} \sum_{i=1}^{n_j} a_{ji} Z_{ji} \right\| \leq C \max \left(\sigma \sqrt{\frac{(t+\log(2m))}{N}}, \frac{K \left(\log \frac{K}{\sigma}\right)^{1/\gamma} (t+\log(2m))}{W} \right). \tag{41}$$

b. For $\mathcal{D}_2$.

Directly using Lemma 2 with $\|\mathbb{E}X_{ji}\| \leq \mu_X, \|\xi_{ji}\|_{L^2} \leq \sigma_\xi$ and $a_{ji} = \frac{\omega_j}{n_j}$, we know that for all $t \geq 0$, with probability at least $1-e^{-t}$

$$\mathcal{D}_2 = \left\| \sum_{j=1}^{T} \sum_{i=1}^{n_j} a_{ji} \mu_X \xi_{ji} \right\| \leq C\mu \max \left(\sigma_\xi \sqrt{\frac{t}{N}}, \frac{K_1 (\log \frac{K_1}{\sigma_\xi})^{1/\alpha} t}{W} \right). \tag{42}$$

c. For $\mathcal{D}_3$. Denote that $Z_{ji} = \langle M_j^0, X_{ji} \rangle X_{ji} - \mathbb{E}(\langle M_j^0, X_{ji} \rangle X_{ji})$ and $a_{ji} = \frac{\omega_j}{n_j}$ with

$$\|Z_{ji}\|_{\psi(\beta/2)} \leq \mathcal{K}, \quad \|Z_{ji}\|_{L^2} \leq \varsigma.$$

Using Lemma 2, we know that for all $t \geq 0$, with probability at least $1-e^{-t}$

$$\mathcal{D}_3 = \left\| \sum_{j=1}^{T} \sum_{i=1}^{n_j} a_{ji} Z_{ji} \right\| \leq C \max \left(\varsigma \sqrt{\frac{(t+\log(2m))}{N}}, \frac{\mathcal{K} \left(\log \frac{\mathcal{K}}{\varsigma}\right)^{2/\beta} (t+\log(2m))}{W} \right). \tag{43}$$

From (41), (42), (43), we know that for all $t > 0$, with probability at least $1-3e^{-t}$,

$$\begin{aligned} \left\| \sum_{j=1}^{T} \omega_j \Delta_j \right\| \leq\ & C_1 \max \left(\sigma \sqrt{\frac{(t+\log(2m))}{N}}, \frac{K \left(\log \frac{K}{\sigma}\right)^{1/\gamma} (t+\log(2m))}{W} \right) \\ & + C_2 \mu_X \max \left(\sigma_\xi \sqrt{\frac{t}{N}}, \frac{K_1 (\log \frac{K_1}{\sigma_\xi})^{1/\alpha} t}{W} \right) \\ & + C_3 \max \left(\varsigma \sqrt{\frac{(t+\log(2m))}{N}}, \frac{\mathcal{K} \left(\log \frac{\mathcal{K}}{\varsigma}\right)^{2/\beta} (t+\log(2m))}{W} \right). \end{aligned}$$

□

Lemma 4 *Under Assumption 4 and 5, if* $h \to 0$ *and* $(m_1 m_2)^{1/2} T h^2 \to \infty$ *as* $m_1, m_2, T \to \infty$, $nTh \gg (K_*/\sigma_*)^2 \log(m_1+m_2)$, *then with probability at least* $1 - 3/(m_1+m_2)$,

$$\mathcal{W}_h \Delta_t \leq C_1 \sigma_* \sqrt{\frac{\log(m_1+m_2)}{nTh}}, \tag{44}$$

where $C_1 > 0$ is a constant independent to $\sigma_*, K_*, T, h, m_1, m_2$ such that

$$\sigma_* = \max\left\{\sigma, \mu_X \sigma_\xi, \varsigma\right\},$$
$$K_* = \max\left\{\mathcal{K}\left(\log\frac{\mathcal{K}}{\varsigma}\right)^{2/\beta}, \mu_X K_1\left(\log\frac{K_1}{\sigma_\xi}\right)^{1/\alpha}, K\left(\log\frac{K}{\sigma}\right)^{1/\gamma}\right\},$$

in which $\mu_X = \max_{i,j}\|\mathbb{E}X_{ji}\|_\infty$, $\sigma_\xi = \max_{i,j}\|\xi_{ji}\|_{L^2}$ and

$$K = \max_{i,j}\|\xi_{ji}(X_{ji} - \mathbb{E}X_{ji})\|_{\psi(\gamma)}, \quad \mathcal{K} = \max_{i,j}\|\langle M_j^0, X_{ji}\rangle X_{ji} - \mathbb{E}(\langle M_j^0, X_{ji}\rangle X_{ji})\|_{\psi(\beta/2)},$$

$$\sigma = \max_{i,j}\|\xi_{ji}(X_{ji} - \mathbb{E}X_{ji})\|_{L^2}, \quad \varsigma = \max_{i,j}\|\langle M_j^0, X_{ji}\rangle X_{ji} - \mathbb{E}(\langle M_j^0, X_{ji}\rangle X_{ji})\|_{L^2},$$

with $\|X\|_{L^2} = \|\mathbb{E}XX^T\|_\infty^{1/2} \vee \|\mathbb{E}X^TX\|_\infty^{1/2}$.

Proof of Lemma 4:

Define that

$$\mathcal{N}_h(t) = \{j : t - Th < j < t + Th\}, \quad \bar{n}_t = \frac{1}{|\mathcal{N}_h(t)|}\sum_{j\in\mathcal{N}_h(t)} n_j, \quad \check{n}_t = \min_{j\in\mathcal{N}_h(t)} n_j.$$

Recall the definition of ω_j that $\omega_j = \omega_h(j-t) = \frac{K\left(\frac{j-t}{Th}\right)}{\sum_{k=1}^T K\left(\frac{k-t}{Th}\right)}$, we have that

$$\begin{aligned} N &= \frac{1}{\sum_{j=1}^T \omega_h(j-t)^2/n_j} = \frac{\left\{\sum_{k=1}^T \frac{1}{Th}K\left(\frac{k-t}{Th}\right)\right\}^2}{\sum_{j=1}^T \frac{1}{T^2h^2}K\left(\frac{j-t}{Th}\right)^2/n_j} \\ &\asymp \frac{1}{\sum_{j=1}^T \frac{1}{n_jT^2h^2}K\left(\frac{j-t}{Th}\right)^2} \asymp \left\{\frac{1}{\bar{n}_tTh}\sum_{j=1}^T \frac{1}{Th}K\left(\frac{j-t}{Th}\right)^2\right\}^{-1} \\ &= \bar{n}_tThR(K)^{-1}, \end{aligned}$$

and

$$\begin{aligned} W &= \frac{1}{\max_j \omega_h(j-t)/n_j} = \left\{\max_j \frac{\frac{1}{Th}K\left(\frac{j-t}{Th}\right)}{n_j\frac{1}{Th}\sum_{k=1}^T K\left(\frac{k-t}{Th}\right)}\right\}^{-1} \\ &\asymp \left\{\max_j \frac{1}{n_jTh}K\left(\frac{j-t}{Th}\right)\right\}^{-1} \asymp \check{n}_tThK(0)^{-1}. \end{aligned}$$

With the assumption $\bar{n}_t, \check{n}_t \asymp n$, for $t \leq \log(m_1+m_2)$, when $nTh \gg \{K_*/\sigma_*\}^2 \log(m_1+m_2)$, the first terms in (41), (42) and (43) dominate those bounds which gives

$$\mathcal{W}_h \Delta_t \leq C_1 \sigma_* \sqrt{\frac{s+\log(m_1+m_2)}{nTh}} \ll 1.$$

By setting $t = \log(m_1+m_2)$, we can deduce Lemma 4 directly.

□

Proof of Theorem 2: With Lemma 4 and Theorem 1, we have

$$\left(\delta_h M_t^2 + 2\mu^{-1}\mathcal{W}_h\Delta_t \left\|M_t^0\right\|_1\right)^{1/2} \gg (1+\sqrt{2})\mu^{-1}\mathcal{W}_h\Delta_t\sqrt{r_t},$$

which leads to the following conclusion. From Lemma 4, we know that for all $t > 0$, with probability at least $1-3e^{-t}$,

$$\left\|\sum_{j=1}^{T}\omega_j\Delta_j\right\|_\infty \leq C_1 \max\left\{\sigma^*\sqrt{\frac{t+\log(m_1+m_2)}{\bar{n}_t Th}}, K^*\frac{t+\log(m_1+m_2)}{\check{n}_t Th}\right\}.$$

Choose that

$$t = \log(m_1+m_2), \quad \lambda = 2\sqrt{2}C_1\sigma^*\sqrt{\frac{\log(m_1+m_2)}{\bar{n}_t Th}}.$$

If $(\check{n}_t^2/\bar{n}_t)Th \gg \{K^*/\sigma^*\}^2\log(m_1+m_2)$, than

$$\sigma^*\sqrt{\frac{t+\log(m_1+m_2)}{\bar{n}_t Th}} \gg K^*\frac{t+\log(m_1+m_2)}{\check{n}_t Th}.$$

From Proposition 1

$$(m_1m_2)^{-1/2}\left\|M_t^0 - \sum_{j=1}^{T}\omega_j M_j^0\right\|_2 \leq \frac{1}{2}\alpha(K)D_2h^2.$$

With Theorem 1, so

$$(m_1m_2)^{-1/2}\left\|\widetilde{M}_t^\lambda - M_t^0\right\|_2 \leq \frac{1}{2}C_2\alpha(K)D_2h^2 + \left(2+\sqrt{2}\right)C_1\sigma^*\left(\frac{r_t\log(m_1+m_2)}{\mu^2 m_1 m_2 \bar{n}_t Th}\right)^{1/2}$$

holds with probability at least $1-3/(m_1+m_2)$.

□

Proof of Corollary 1: Plug in the upper bound of $\mathcal{W}_h\Delta_t$ in Lemma 4, Corollary 1 can be obtained immediately. □

Proof of Corollary 2:

From the distribution of X_{ji} and Assumption 4, we have that

$$\gamma = \alpha, \quad \beta = \infty, \quad \mu = (m_1 m_2)^{-1},$$

and

$$\sigma \leq \sigma_\xi \sqrt{\frac{1}{m_1 \wedge m_2}}, \quad \mu_X \sigma_\xi = \sigma_\xi \sqrt{\frac{1}{m_1 \wedge m_2}}, \quad \varsigma \leq M \sqrt{\frac{1}{m_1 \wedge m_2}},$$
$$K \asymp \sigma_\xi, \quad K_1 \asymp \sigma_\xi, \quad \mathcal{K} \asymp M.$$

Then it can be obtained that

$$\sigma^* = (C_M \vee \sigma_\xi) \sqrt{\frac{1}{m_1 \wedge m_2}},$$
$$K^*/\sigma^* \asymp (m_1 \wedge m_2)^{1/2} \log^{1/\alpha}(m_1 \wedge m_2).$$

Then $(K^*/\sigma^*)^2 \log(m_1+m_2) \asymp (m_1 \wedge m_2) \log^{2/\alpha}(m_1 \wedge m_2) \log(m_1+m_2)$. Note that $\log(m_1+m_2) > \log(m_1 \wedge m_2)$, when $(\check{n}_t^2/\bar{n}_t)Th \gg (m_1 \wedge m_2) \log^{1+2/\alpha}(m_1+m_2)$, the first term in (44) dominates the bound. Apply Theorem 2 and Corollary 1 and the proof is finished.

□

8 Heterogeneous Scenario

Consider the heterogeneous assumption first.

Assumption 7 *(Heterogeneous Assumption) The second moment matrix Σ_t of X_t is positive definite with smallest eigenvalue $\mu_t > 0$ for $t = 1, 2, \cdots, T$.*

Here we provide the corresponding results of Theorem 1, Theorem 2 and Corollary 1 under the heterogeneous scenario. The key difference between homogeneous and heterogeneous scenarios is that the covariance matrix used for time points t is Σ in homogeneous case and $\sum_{j=1}^{T}\omega_h(j-t)\Sigma_j$ in heterogeneous case. Define $\nu_t=\sum_{j=1}^{T}\omega_h(j-t)\mu_t$, and the difference is using ν_t to replace μ in each theorems and corollaries for the upper bound of $\widetilde{M}_t^{\lambda}$.

Theorem 6 *(Heterogeneous Theorem 1) Under Assumption 7, if* $\lambda \geq 2\mathcal{W}_h\Delta_t$, *then*

$$\left\|\widetilde{M}_t^{\lambda}-M_t^0\right\|_2 \leq \delta_h M_t+\min\left\{\left((\delta_h M_t)^2+\frac{2\lambda}{\nu_t}\left\|M_t^0\right\|_1\right)^{1/2},\frac{1+\sqrt{2}}{2}\frac{\lambda}{\nu_t}\sqrt{r_t}\right\}. \tag{45}$$

When selecting $\lambda=2\mathcal{W}_h\Delta_t$, *we have*

$$\left\|\widetilde{M}_t^{\lambda}-M_t^0\right\|_2 \leq \delta_h M_t+\min\left\{\left((\delta_h M_t)^2+2\nu_t^{-1}\mathcal{W}_h\Delta_t\left\|M_t^0\right\|_1\right)^{1/2},(1+\sqrt{2})\nu_t^{-1}\mathcal{W}_h\Delta_t\sqrt{r_t}\right\}. \tag{46}$$

Proof In the Proof of Theorem 1, the whole proof is not changed but changing (28) as

$$\begin{aligned}&\left(\operatorname{vec}(\widehat{M}_t^{\lambda})-\operatorname{vec}(M_t^0)\right)^{\top}\Sigma\left[\left(\operatorname{vec}(\widehat{M}_t^{\lambda})-\operatorname{vec}(M_t^0)\right)+\left(2\operatorname{vec}(M_t^0)-2\sum_{j=1}^{T}\omega_j\operatorname{vec}(M_j^0)\right)\right]\\ \geq&\nu_t\left\|\widetilde{M}_t^{\lambda}-M_t^0\right\|_2^2-\nu_t\left\|\widetilde{M}_t^{\lambda}-M_t^0\right\|_2\left\|M_t^0-\sum_{j=1}^{T}\omega_j M_j^0\right\|_2.\end{aligned}$$

and changing (35) as

$$\begin{aligned}&\sum_{j=1}^{T}\omega_j\left[\mathcal{C}_j(\widetilde{M}_t^{\lambda}-M_t^0,\widetilde{M}_t^{\lambda}-M_t^0)+\mathcal{C}_j(\widetilde{M}_t^{\lambda}-M_t^0,M_t^0-M_j^0)\right]\\ =&\left(\operatorname{vec}(\widehat{M}_t^{\lambda})-\operatorname{vec}(M_t^0)\right)^{\top}\Sigma\left(\left(\operatorname{vec}(\widehat{M}_t^{\lambda})-\operatorname{vec}(M_t^0)\right)+\left(M_t^0-\sum_{j=1}^{T}\omega_j M_j^0\right)\right)\\ \geq&\nu_t\left\|\widetilde{M}_t^{\lambda}-M_t^0\right\|_2^2-\nu_t\left\|\widetilde{M}_t^{\lambda}-M_t^0\right\|_2\left\|M_t^0-\sum_{j=1}^{T}\omega_j M_j^0\right\|_2.\end{aligned}$$

□

Theorem 7 *(Heterogeneous Theorem 2) Under Assumption 1-5, let* $nTh \gg (K_*/\sigma_*)^2 \log(m_1+m_2)$, $h \to 0$ *and* $(m_1 m_2)^{1/2} Th^2 \to \infty$ *as* $n, m_1, m_2, T \to \infty$, *when*

$$\lambda = 2C_1\sigma_* \sqrt{\frac{\log(m_1+m_2)}{nTh}},$$

where C_1 *and* σ_* *are defined in Lemma 4 of Supplementary Material, then with probability at least* $1 - 3/(m_1+m_2)$,

$$(m_1 m_2)^{-1/2} \left\| \widetilde{M}_t^{\lambda} - M_t^0 \right\|_2 \leq \frac{1}{2}\alpha(K) D_2 h^2 + \left(1+\sqrt{2}\right) C_1 \sigma_* \left(\frac{r_t \log(m_1+m_2)}{\nu_t^2 m_1 m_2 nTh} \right)^{1/2} + o(h^2), \tag{47}$$

where $\alpha(K)$ *is defined in (10) and* D_2 *in Assumption 2.*

Corollary 5 *(Heterogeneous Corollary 1) Under assumptions of Theorem 7, when*

$$nT \gg \max\{K_*^{5/2}\sigma_*^{-3}(\nu_t^2 m_1 m_2)^{1/4} \log(m_1+m_2),\ \sigma_*^2(\nu_t^2 m_1 m_2)^{-1} \log(m_1+m_2)\},$$

with probability at least $1 - 3/(m_1+m_2)$,

$$(m_1 m_2)^{-1/2} \left\| \widetilde{M}_t^{\lambda} - M_t^0 \right\|_2 \leq C_2 \left(\frac{\sigma_*^2 r_t \log(m_1+m_2)}{\nu_t^2 m_1 m_2 nT} \right)^{2/5}, \tag{48}$$

where $C_2 = 1/2 \left[(2+2\sqrt{2})C_1\right]^{4/5} \left[\alpha(K) D_2\right]^{1/5}$.

9 Definition of $\phi-$mixing

The dependence of two σ-field $\mathcal{A}$ and $\mathcal{B}$ is measured by

$$\phi(\mathcal{A}, \mathcal{B}) = \sup \left\{ \left| P(A) - \frac{P(A \cap B)}{P(B)} \right| ; A \in \mathcal{A}, B \in \mathcal{B}, P(B) \neq 0 \right\}.$$

And for a sequence of σ-field $\mathcal{A}_j, j \in \mathbb{N}^*$, the ϕ-coefficients $\phi_{\mathcal{A}}(k), k \in \mathbb{N}$ is defined as

$$\phi_{\mathcal{A}}(k) = \sup_{|j-s| \leq k} \phi(\mathcal{A}_j, \mathcal{A}_s).$$

If $\lim_{k \to \infty} \phi_{\mathcal{A}}(k) = 0$, the sequence of σ-field $\mathcal{A}_j, j \in \mathbb{N}^*$ is said to be ϕ-mixing.

10 Proofs of Theorem 3 and Corollary 3, 6

Lemma 5 *Under Assumption 4 and 6, if* $h \to 0$ *as* $n, T \to \infty$, $Th \gg \log(m_1 + m_2)$ *and* $nTh \gg (K_*/\sigma_*)^2 \log^3(m_1 + m_2)$, *with probability at least* $1 - 3/(m_1 + m_2)$,

$$\mathcal{W}_h \Delta_t \leq \mathcal{C}_1 (\Phi_{\mathcal{X}} \vee \Phi_{\mathcal{Y}}) \sigma_* \sqrt{\frac{\log(m_1 + m_2)}{nTh}},$$

where $\mathcal{C}_1 > 0$ *is a constant independent to* $\sigma_*, K_*, T, h, m_1, m_2$.

Proof of Lemma 5 and Theorem 3: Denote $\mathbf{X}_j = (X_{j1}, X_{j2}, \ldots, X_{jn_j})^\top, \xi_j = (\xi_{j1}, \xi_{j2}, \ldots, \xi_{jn_j})^\top$. We know that $\sigma(\mathbf{X}_j - \mathbb{E}\mathbf{X}_j) \subseteq \mathcal{X}_j$. Define $\mathcal{Z}_j$ such that $\mathcal{Z}_j = \sigma((\mathbf{X}_j^\top, \xi_j^\top)^\top)$, then

$$\begin{aligned}
&\left| P(A_x \otimes A_y) - \frac{P(A_x \otimes A_y \cap B_x \otimes B_y)}{P(B_x \otimes B_y)} \right| \\
=&\left| P(A_x) \left(P(A_y) - \frac{P(A_y \cap B_y)}{P(B_y)} \right) + \frac{P(A_y \cap B_y)}{P(B_y)} \left(P(A_x) - \frac{P(A_x \cap B_x)}{P(B_x)} \right) \right| \\
\leq&\phi_{\mathcal{X}}(|j-l|) + \phi_{\mathcal{Y}}(|j-l|) - \phi_{\mathcal{X}}(|j-l|)\phi_{\mathcal{Y}}(|j-l|),
\end{aligned}$$

where $A_x \otimes A_y \in \mathcal{Z}_j,\ B_x \otimes B_y \in \mathcal{Z}_l$. So we know that $\phi_{\mathcal{Z}} \leq \phi_{\mathcal{X}} + \phi_{\mathcal{Y}} - \phi_{\mathcal{X}}\phi_{\mathcal{Y}}$, which means that

$$\Phi_{\mathcal{Z}} = \sum_{k=1}^{\infty} \sqrt{\phi_{\mathcal{Z}}(k)} \leq \Phi_{\mathcal{X}} + \Phi_{\mathcal{Y}} < \infty.$$

Let $Z_{ji} = \frac{\omega_j}{n_j} \xi_{ji} (X_{ji} - \mathbb{E}X_{ji})$ and $Z_j = \sum_{i=1}^{n_j} Z_{ji}$. Using Lemma 2, we know that

$$P(\|Z_j\| \geq s) \leq 2m \exp\left\{ -C \min\left\{ \frac{n_j s^2}{w_j^2 \sigma^2}, \frac{n_j s}{\omega_j K (\log \frac{K}{\sigma})^{1/\gamma}} \right\} \right\}.$$

Denote the bounded function

$$g((\mathbf{X}_j^\top, \xi_j^\top)^\top) = \frac{\omega_j}{n_j} (\mathbf{X}_j - \mathbb{E}\mathbf{X}_j)^\top \xi_j \mathbf{1}_{\left\| \frac{\omega_j}{n_j} (\mathbf{X}_j - \mathbb{E}\mathbf{X}_j)^\top \xi_j \right\| \leq s},$$

we know that

$$P\left(\left\|\sum_{j=1}^{T} Z_j\right\| \geq t\right) \leq \sum_{j=1}^{T} P(\|Z_j\| \geq s) + P\left(\left\|\sum_{j=1}^{T} Z_j \mathbf{1}_{\|Z_j\| \leq s}\right\| \geq t\right)$$
$$= \sum_{j=1}^{T} P(\|Z_j\| \geq s) + P\left(\left\|\sum_{j=1}^{T} g((\mathbf{X}_j^\top, \xi_j^\top)^\top)\right\| \geq t\right).$$

With the tail distribution assumptions for X_j, ξ_j, when $s \leq \frac{\omega_j \sigma^2}{K(\log \frac{K}{\sigma})^{1/\gamma}}$, we have the inequalities

$$\left\|\mathbb{E}\sum_{j=1}^{T} g((\mathbf{X}_j^\top, \xi_j^\top)^\top)\right\| = \left\|\sum_{j=1}^{T} Z_j \mathbf{1}_{\|Z_j\| \geq s}\right\|$$
$$\leq \sum_{j=1}^{T} \left| \int_s^\infty P(\|Z_j\| \geq s) dx + sP(\|Z_j\| \geq s) \right|$$
$$\leq \left(s + \frac{c}{snT^2h^2}\right) \sum_{j=1}^{T} P(\|Z_j\| \geq s).$$

And when $s \geq \frac{\omega_j \sigma^2}{K(\log \frac{K}{\sigma})^{1/\gamma}}$, we have that

$$\left\|\mathbb{E}\sum_{j=1}^{T} g((\mathbf{X}_j^\top, \xi_j^\top)^\top)\right\| \leq \left(s + \frac{cK(\log \frac{K}{\sigma})^{1/\gamma}}{nTh}\right) \sum_{j=1}^{T} P(\|Z_j\| \geq s).$$

Meanwhile we have the bound

$$\left\|\mathbb{E}\sum_{j=1}^{T} g^2((\mathbf{X}_j^\top, \xi_j^\top)^\top)\right\| \leq \sum_{j=1}^{T} \frac{\omega_j^2}{n_j} \sigma_\xi^2 \sigma_X^2 = \frac{\sigma^2}{N}.$$

Similar to Theorem 3 in Samson (2000), with the matrix value function $g(\cdot)$ and norm $\|\cdot\|_\infty$

$$P\left(\left\|\sum_{j=1}^{T} g((\mathbf{X}_j^\top, \xi_j^\top)^\top)\right\| \geq t\right)$$
$$\leq P\left(\left\|\sum_{j=1}^{T} g((\mathbf{X}_j^\top, \xi_j^\top)^\top) - \mathbb{E}\sum_{j=1}^{T} g((\mathbf{X}_j^\top, \xi_j^\top)^\top)\right\| \geq t - \left\|\mathbb{E}\sum_{j=1}^{T} g((\mathbf{X}_j^\top, \xi_j^\top)^\top)\right\|\right)$$
$$\leq 2m \exp\left\{-\frac{1}{8\Phi_Z^2} \min\left\{\frac{t}{2s}, \frac{Nt^2}{16\sigma^2}\right\}\right\},$$

when $t \geq \left\| \mathbb{E} \sum_{j=1}^{T} g((\mathbf{X}_j^\top, \xi_j^\top)^\top) \right\|$.

Thus we have

$$
\begin{aligned}
& P\left(\left\| \sum_{j=1}^{T} Z_j \right\| \geq t \right) \\
\leq & \sum_{j=(t-h/2)\vee 0}^{(t+h/2)\wedge T} 2m \exp\left\{ -C \min\left\{ \frac{n_j s^2}{w_j^2 \sigma^2}, \frac{n_j s}{\omega_j K (\log \frac{K}{\sigma})^{1/\gamma}} \right\} \right\} \\
& + 2m \exp\left\{ -\frac{1}{8\Phi_{\mathcal{Z}}^2} \min\left\{ \frac{t}{2s}, \frac{Nt^2}{16\sigma^2} \right\} \right\} \\
\leq & \sum_{j=(t-h/2)\vee 0}^{(t+h/2)\wedge T} 2m \exp\left\{ -\frac{C n_j s^2}{\omega_j^2 \sigma^2} \right\} + 2m \exp\left\{ -\frac{1}{8\Phi_{\mathcal{Z}}^2} \min\left\{ \frac{t}{2s}, \frac{Nt^2}{16\sigma^2} \right\} \right\} \\
\lesssim & 2m \exp\left\{ -\frac{CnT^2h^2s^2}{\sigma^2} + \log Th \right\} + 2m \exp\left\{ -\frac{1}{8\Phi_{\mathcal{Z}}^2} \min\left\{ \frac{t}{2s}, \frac{Nt^2}{16\sigma^2} \right\} \right\}
\end{aligned}
$$

when $t \geq \left\| \mathbb{E} \sum_{j=1}^{T} g((\mathbf{X}_j^\top, \xi_j^\top)^\top) \right\|$ and $s \leq \frac{\omega_j \sigma^2}{K(\log \frac{K}{\sigma})^{1/\gamma}}$.

And similarly

$$
\begin{aligned}
& P\left(\left\| \sum_{j=1}^{T} Z_j \right\| \geq t \right) \\
\lesssim & \, 2m \exp\left\{ -C \frac{nThs}{K(\log \frac{K}{\sigma})^{1/\gamma}} + \log Th \right\} + 2m \exp\left\{ -\frac{1}{8\Phi_{\mathcal{Z}}^2} \min\left\{ \frac{t}{2s}, \frac{Nt^2}{16\sigma^2} \right\} \right\}
\end{aligned}
$$

when $t \geq \left\| \mathbb{E} \sum_{j=1}^{T} g((\mathbf{X}_j^\top, \xi_j^\top)^\top) \right\|$ and $s \geq \frac{\omega_j \sigma^2}{K(\log \frac{K}{\sigma})^{1/\gamma}}$.

When

$$
n \gtrsim \left(K(\log K/\sigma)^{1/\gamma}/\sigma \right)^2 \log(Thm),
$$

choose

$$
s \asymp n^{-1/2}(Th)^{-1}\sigma \log^{1/2}(Thm), \quad t \asymp n^{-1/2}(Th)^{-1/2}\sigma\Phi_{\mathcal{Z}} \log^{1/2} m.
$$

If

$$
Th \gg \Phi_{\mathcal{Z}}^2 \log(Thm) \log m,
$$

we have that $t \gg s\Phi_{\mathcal{Z}}^2 \log m$.

Meanwhile, when

$$
n \lesssim \left(K(\log K/\sigma)^{1/\gamma}/\sigma \right)^2 \log(Thm),
$$

choose

$$s \asymp n^{-1}(Th)^{-1}K(\log K/\sigma)^{1/\gamma}\log(Thm), \quad t \asymp n^{-1/2}(Th)^{-1/2}\sigma\Phi_{\mathcal{Z}}\log^{1/2} m.$$

If

$$nTh \gg (K(\log K/\sigma)^{1/\gamma}/\sigma)^2\Phi_{\mathcal{Z}}^2\log^2(Thm)\log m,$$

we have that $t \gg s\Phi_{\mathcal{Z}}^2\log m$.

Thus when

$$Th \gg \Phi_{\mathcal{Z}}^2\log(Thm)\log m$$

and

$$nTh \gg (K(\log K/\sigma)^{1/\gamma}/\sigma)^2\Phi_{\mathcal{Z}}^2\log^2(Thm)\log m,$$

the inequality

$$P\left(\left\|\sum_{j=1}^{T}\sum_{i=1}^{n_j}\frac{\omega_j}{n_j}\xi_{ji}(X_{ji}-\mathbb{E}(X_{ji}))\right\| \geq t\right) \leq \frac{1}{m}.$$

holds where $t \asymp n^{-1/2}(Th)^{-1/2}\sigma\Phi_{\mathcal{Z}}\log^{1/2} m$.

Similarly, we choose $t \asymp n^{-1/2}(Th)^{-1/2}\sigma_\xi\mu_X\Phi_{\mathcal{Y}}\log^{1/2} m$ and

$$P\left(\left\|\sum_{j=1}^{T}\sum_{i=1}^{n_j}\frac{\omega_j}{n_j}\xi_{ji}\mathbb{E}(X_{ji})\right\| \geq t\right) \leq \frac{1}{m}$$

holds when $Th \gg \Phi_{\mathcal{Y}}^2\log(Thm)\log m$ and

$$nTh \gg (K_1(\log K_1/\sigma_\xi)^{1/\alpha}/\sigma_\xi)^2\Phi_{\mathcal{Y}}^2\log^2(Thm)\log m.$$

And choose $t \asymp n^{-1/2}(Th)^{-1/2}\varsigma\Phi_{\mathcal{X}}\log^{1/2} m$, we have

$$P\left(\left\|\sum_{j=1}^{T}\sum_{i=1}^{n_j}\frac{\omega_j}{n_j}\left\{\langle M_0(j), X_{ji}\rangle X_{ji} - E\left(\langle M_0(j), X_{ji}\rangle X_{ji}\right)\right\}\right\| \geq t\right) \leq \frac{1}{m}$$

holds when $Th \gg \Phi_{\mathcal{X}}^2\log(Thm)\log m$ and $nTh \gg (\mathcal{K}(\log\mathcal{K}/\varsigma)^{2/\beta}/\varsigma)^2\Phi_{\mathcal{X}}^2\log^2(Thm)\log m$.

Summarizing the above results with $\Phi_{\mathcal{Z}} \leq \Phi_{\mathcal{X}} + \Phi_{\mathcal{Y}}$, we know that when $Th \gg (\Phi_{\mathcal{X}}^2 \vee \Phi_{\mathcal{Y}}^2)\log(Thm)$ and $nTh \gg (K^*/\sigma^*)^2(\Phi_{\mathcal{X}}^2 \vee \Phi_{\mathcal{Y}}^2)\log^2(Thm)\log m$, there exists a constant

$\mathcal{C}_1 > 0$, with probability at least $1 - 3/m$,

$$\left\| \sum_{j=1}^{T} \omega_j \Delta_j \right\| \leq \mathcal{C}_1 (\Phi_{\mathcal{X}} \vee \Phi_{\mathcal{Y}}) \sigma^* \sqrt{\frac{\log(m_1 + m_2)}{nTh}}.$$

Choose

$$\lambda = 2\mathcal{C}_1 (\Phi_{\mathcal{X}} \vee \Phi_{\mathcal{Y}}) \sigma^* \sqrt{\frac{\log(m_1 + m_2)}{nTh}}$$

and using Theorem 1 and proof of Theorem 2, we have that

$$(m_1 m_2)^{-1/2} \left\| \widetilde{M}_t^{\lambda} - M_t^0 \right\|_2 \leq \frac{1}{2} \alpha(K) D_2 h^2 + (1 + \sqrt{2}) \mathcal{C}_1 (\Phi_{\mathcal{X}} \vee \Phi_{\mathcal{Y}}) \sigma^* \left(\frac{r_t \log(m_1 + m_2)}{\mu^2 m_1 m_2 nTh} \right)^{1/2}.$$

Note that $Th \gg \log(Th)$ when $Th \to \infty$, we can represent the condition $Th \gg (\Phi_{\mathcal{X}}^2 \vee \Phi_{\mathcal{Y}}^2) \log(Thm)$ and $nTh \gg (K^*/\sigma^*)^2 (\Phi_{\mathcal{X}}^2 \vee \Phi_{\mathcal{Y}}^2) \log^2(Thm) \log m$ as $Th \gg (\Phi_{\mathcal{X}}^2 \vee \Phi_{\mathcal{Y}}^2) \log m$ and $nTh \gg (K^*/\sigma^*)^2 (\Phi_{\mathcal{X}}^2 \vee \Phi_{\mathcal{Y}}^2) \log^3 m$ respectively.

□

Corollary 6 *Under assumptions of Theorem 3, when X_{ji} are i.i.d. uniformly distributed on $\mathcal{E}$, ξ_{ji} are independently follow sub-exponential mean-zero distributions, $Th \gg \log(m_1 + m_2)$ and $nTh \gg (m_1 \wedge m_2) \log^{3+2/\alpha}(m_1 + m_2)$, then with probability at least $1 - 3/(m_1 + m_2)$,*

$$(m_1 m_2)^{-1/2} \left\| \widetilde{M}_t^{\lambda} - M_t^0 \right\|_2 \leq \frac{1}{2} \alpha(K) D_2 h^2 + (1 + \sqrt{2}) \mathcal{C}_1 (\Phi_{\mathcal{X}} \vee \Phi_{\mathcal{Y}}) (C_M \vee \sigma_\xi) \left(\frac{r_t (m_1 \vee m_2) \log(m_1 + m_2)}{nTh} \right)^{1/2}.$$

When $nT \gg (m_1 \vee m_2) \log^{(1+5/2\alpha)}(m_1 + m_2)$ and $n^{-1} T^4 \gg (m_1 \vee m_2)^{-1} \log^4(m_1 + m_2)$, we select

$$h = \mathcal{C}_h \left(\frac{(C_M \vee \sigma_\xi)^2 (\Phi_{\mathcal{X}} \vee \Phi_{\mathcal{Y}})^2 r_t (m_1 \vee m_2) \log(m_1 + m_2)}{nT} \right)^{1/5},$$

then

$$(m_1 m_2)^{-1/2} \left\| \widetilde{M}_t^{\lambda} - M_t^0 \right\|_2 \leq \mathcal{C}_2 (\Phi_{\mathcal{X}} \vee \Phi_{\mathcal{Y}})^{4/5} (C_M \vee \sigma_\xi)^{4/5} \left(\frac{r_t (m_1 \vee m_2) \log(m_1 + m_2)}{nT} \right)^{2/5}, \tag{49}$$

where $\mathcal{C}_1, \mathcal{C}_h, \mathcal{C}_2$ are the same constants as above.

The proofs of Corollary 3 and 6 are similar to those of Corollary 1 and 2, which we do not repeat.

11 Application to Compressed Sensing

Corollary 7 *(Independent case) Under Assumption 1-5, when X_{ji} are random matrices with independent mean-zero sub-gaussian elements with variance σ_X^2, ξ_{ji} are independently follow sub-gaussian distributions, $nTh \gg m_1 \vee m_2$, $h \to 0$ and $(m_1 m_2)^{1/2} T h^2 \to \infty$ as $n, m_1, m_2, T \to \infty$, with probability at least $1 - 2/(m_1 + m_2)$,*

$$
\begin{aligned}
&(m_1 m_2)^{-1/2} \left\| \widetilde{M}_t^{\lambda} - M_t^0 \right\|_2 \\
&\leq \frac{1}{2} \alpha(K) D_2 h^2 + \left(1 + \sqrt{2}\right) C_1 \left[\eta \vee \{ M (m_1 m_2)^{1/2} \} \right] \left(\frac{r_t \log(m_1 + m_2)}{(m_1 \wedge m_2) n T h} \right)^{1/2} + o(h^2),
\end{aligned}
$$

When $nT \gg (m_1 \vee m_2) \log(m_1 + m_2)$, let

$$
h = C_h \left(\frac{\{ \eta^2 \vee (M^2 m_1 m_2) \} \, r_t \log(m_1 + m_2)}{(m_1 \wedge m_2) n T} \right)^{1/5},
$$

then

$$
(m_1 m_2)^{-1/2} \left\| \widetilde{M}_t^{\lambda} - M_t^0 \right\|_2 \leq C_2 \left(\frac{\{ \eta^2 \vee (M^2 m_1 m_2) \} \, r_t \log(m_1 + m_2)}{(m_1 \wedge m_2) n T} \right)^{2/5}, \tag{50}
$$

where $\eta = \sigma_\xi / \sigma_X$ and C_1, C_h, C_2 are the same constants as above.

Theorem 6 in Koltchinskii (2011b) presented the error bound for static matrix compressed sensing which is

$$
(m_1 m_2)^{-1/2} \left\| \widehat{M}_t^{\lambda} - M_t^0 \right\|_2 \leq C_4 \eta \left(\frac{r_t}{(m_1 \wedge m_2) n} \right)^{1/2}.
$$

Similarly to the matrix completion, when $T \gg \eta^{-1} \{ \eta^2 \vee (M^2 m_1 m_2) \} \{ n (m_1 \wedge m_2) \}^{1/4} \log(m_1 + m_2)$, our dynamic method gives a sharper bound than the static method.

From Corollary 2 and Corollary 7, when the variances of observation errors ξ_{ji} are small enough comparing to the variances of X_{ji}, i.e., $\sigma_\xi \leq M$ in matrix completion setting and $\eta \leq M(m_1 m_2)^{1/2}$ in compressed sensing setting, the upper bound (15) and (50) have the same order such that

$$(m_1 m_2)^{1/2} \left\| \widetilde{M}_t^\lambda - M_t^0 \right\|_2 \leq C_2 M^{4/5} \left(\frac{r_t(m_1 \vee m_2) \log(m_1 + m_2)}{nT} \right)^{2/5}.$$

Proof of Corollary 7: Because $\mathbb{E}X_{ji} = 0$, we know that $\mu_X = 0$ and (b) in (40) becomes zero. From the distribution of X_{ji} and Assumption 4, we know that $\beta = 2$ and $K_2 \asymp (m_1 \vee m_2)^{1/2}\sigma_X$ by Theorem 4.4.5 in Vershynin (2018). Also, it is easy to check that $K \asymp \sigma_\xi \sigma_X (m_1 \vee m_2)^{1/2},\ \sigma = \sigma_\xi \sigma_X (m_1 \vee m_2)^{1/2}$. With that

$$\begin{aligned}
&\left\| \mathbb{E}(\langle M_j^0, X_{ji}\rangle X_{ji} - \mathbb{E}(\langle M_j^0, X_{ji}\rangle X_{ji})(\langle M_j^0, X_{ji}\rangle X_{ji} - \mathbb{E}(\langle M_j^0, X_{ji}\rangle X_{ji})^T \right\|_\infty \\
\leq & M^2 m_1 m_2 (m_1 \vee m_2) \sigma_X^4
\end{aligned}$$

and with Lemma 3

$$\begin{aligned}
\left\| \langle M_j^0, X_{ji}\rangle X_{ji} - \mathbb{E}\langle M_j^0, X_{ji}\rangle X_{ji} \right\|_{\psi(1)} &\asymp \left\| \langle M_j^0, X_{ji}\rangle X_{ji} \right\|_{\psi(1)} \\
&\leq \left\| \langle M_j^0, X_{ji}\rangle \right\|_{\psi(2)} \left\| X_{ji} \right\|_{\psi(2)} \\
&\leq M (m_1 m_2)^{1/2} \sigma_X^2,
\end{aligned}$$

we have that $\varsigma \leq M(m_1 m_2)^{1/2}(m_1 \vee m_2)^{1/2}\sigma_X^2, \mathcal{K} \lesssim M(m_1 m_2)^{1/2}\sigma_X^2$.

Then it can be obtained that

$$\begin{aligned}
\sigma^* &= \left[\sigma_\xi \vee M\sigma_X (m_1 m_2)^{1/2} \right] (m_1 \vee m_2)^{1/2} \sigma_X \\
K^*/\sigma^* &\asymp (m_1 \vee m_2)^{-1/2} \log(1/(m_1 \vee m_2)).
\end{aligned}$$

Apply Theorem 2 and Corollary 1, the proof is completed. □

Corollary 8 *(Dependent case.) Under assumptions of Theorem 3, when X_{ji} are random matrices with independent mean-zero sub-gaussian elements with variance σ_X^2 and ξ_{ji} are independently follow sub-gaussian distributions, $Th \gg \log(m_1+m_2)$ and $nTh \gg (m_1 \vee m_2)$, with probability at least $1-2/(m_1+m_2)$,*

$$
\begin{aligned}
&(m_1 m_2)^{-1/2} \left\| \widetilde{M}_t^{\lambda} - M_t^0 \right\|_2 \\
&\leq \frac{1}{2}\alpha(K) D_2 h^2 + \left(1+\sqrt{2}\right) \mathcal{C}_1 (\Phi_{\mathcal{X}} \vee \Phi_{\mathcal{Y}}) \left[\eta \vee M (m_1 m_2)^{1/2}\right] \left(\frac{r_t \log(m_1+m_2)}{(m_1 \wedge m_2) nTh} \right)^{1/2}.
\end{aligned}
$$

When $nT \gg (m_1 \vee m_2)\log(m_1+m_2)$ and $n^{-1}T^4 \gg (m_1 \vee m_2)^{-1}\log^4(m_1+m_2)$, we select

$$
h = \mathcal{C}_h \left(\frac{(\eta^2 \vee M^2 m_1 m_2)\,(\Phi_{\mathcal{X}} \vee \Phi_{\mathcal{Y}})^2 r_t \log(m_1+m_2)}{(m_1 \wedge m_2) nT} \right)^{1/5},
$$

then

$$
(m_1 m_2)^{1/2} \left\| \widetilde{M}_t^{\lambda} - M_t^0 \right\|_2 \leq \mathcal{C}_2 (\Phi_{\mathcal{X}} \vee \Phi_{\mathcal{Y}})^{4/5} \left(\frac{(\eta^2 \vee M^2 m_1 m_2)\, r_t \log(m_1+m_2)}{(m_1 \wedge m_2) nT} \right)^{2/5},
$$

where $\eta = \sigma_\xi / \sigma_X$ and $\mathcal{C}_1, \mathcal{C}_h, \mathcal{C}_2$ are the same constants as above.

The proof of Corollary 8 is similar to that of Corollary 7, which we do not repeat.

12 Proof of Theorem 4

Proof of Theorem 4: First we know that

$$
\nabla f_t(M) = 2 \sum_{j=1}^{T} \omega_j \left[\frac{1}{n_j} \sum_{i=1}^{n_j} \mathbb{E}\langle M, X_{ji} \rangle X_{ji} - \frac{1}{n_j} \sum_{i=1}^{n_j} Y_{ji} X_{ji} \right].
$$

This gives

$$
\begin{aligned}
\|\nabla f_t(M) - \nabla f_t(N)\|_2 &= \left\| 2 \sum_{j=1}^{T} \frac{\omega_j}{n_j} \sum_{i=1}^{n_j} \mathbb{E}\langle M-N, X_{ji} \rangle X_{ji} \right\|_2 \\
&\leq 2 \left\| \sum_{j=1}^{T} \frac{\omega_j}{n_j} \sum_{i=1}^{n_t} (\|X_{ji}\|_2 X_{ji}) \right\|_2 \|M-N\|_2.
\end{aligned}
$$

Define that $2L_f = 4\left\|\sum_{j=1}^{T}\omega_h(j-t)/n_j\sum_{i=1}^{n_j}(\|X_{ji}\|_2 X_{ji})\right\|_2$, Toh & Yun (2010) proved that for any $k>1$,

$$F_t(M_t^{(k)}) - F_t(\widetilde{M}_t^{\lambda}) \le \frac{2L_f\|M_t^{(0)} - \widetilde{M}_t^{\lambda}\|_2^2}{(k+1)^2}. \tag{51}$$

Note that (51) focuses on the convergence rate of object function $F_t(M_t^{(k)})$ instead of $M_t^{(k)}$.from (51) we know

$$\begin{aligned}
&\sum_{j=1}^{T}\omega_j\mathcal{C}_j(M_t^{(k)}, M_t^{(k)}) - 2\sum_{j=1}^{T}\omega_j\left\langle\frac{1}{n_j}\sum_{i=1}^{n_j}Y_{ji}X_{ji}, M_t^{(k)}\right\rangle + \lambda\|M_t^{(k)}\|_1 \\
\le&\sum_{j=1}^{T}\omega_j\mathcal{C}_j(\widehat{M}_t^{\lambda}, \widehat{M}_t^{\lambda}) - 2\sum_{j=1}^{T}\omega_j\left\langle\frac{1}{n_j}\sum_{i=1}^{n_j}Y_{ji}X_{ji}, \widehat{M}_t^{\lambda}\right\rangle + \lambda\|\widehat{M}_t^{\lambda}\|_1 + \frac{2L_f\gamma_t^2}{(k+1)^2} \\
\le&\sum_{j=1}^{T}\omega_j\mathcal{C}_j(M, M) - 2\sum_{j=1}^{T}\omega_j\left\langle\frac{1}{n_j}\sum_{i=1}^{n_j}Y_{ji}X_{ji}, M\right\rangle + \lambda\|M\|_1 + \frac{2L_f\gamma_t^2}{(k+1)^2}
\end{aligned}$$

for any $M\in\mathbb{M}$. Therefore, we have

$$\begin{aligned}
&\sum_{j=1}^{T}\omega_j\mathcal{C}_j(M_t^{(k)} - M_j^0, M_t^{(k)} - M_j^0) \\
\le&\sum_{j=1}^{T}\omega_j\mathcal{C}_j(M - M_j^0, M - M_j^0) + \left\langle 2\sum_{j=1}^{T}\omega_j\Delta_j, M_t^{(k)} - M\right\rangle \\
&+\lambda(\|M\|_1 - \|M_t^{(k)}\|_1) + \frac{2L_f\gamma_t^2}{(k+1)^2}.
\end{aligned}$$

With the assumption that $\lambda > 2\|\sum_{j=1}^{T}\omega_j\Delta_j\|_\infty$, we use the similar methods in proof of Theorem 1 and immediately know that

$$\begin{aligned}
&\left\|M_t^{(k)} - M_t^0\right\|_2 \\
\le&\left\|M_t^0 - \sum_{j=1}^{T}\omega_h(j-t)M_j^0\right\|_2 + \left(\left\|M_t^0 - \sum_{j=1}^{T}\omega_h(j-t)M_j^0\right\|_2^2 + \frac{2\lambda}{\mu}\|M_t^0\|_1 + \frac{2L_f\gamma_t^2}{\mu(k+1)^2}\right)^{1/2}.
\end{aligned}$$

Next, we consider $F_t(\widetilde{M}_t^{\lambda} + M) - F_t(\widetilde{M}_t^{\lambda})$ with the decomposition

$$\begin{aligned}
M &= P_{S_1^{\perp}}MP_{S_2^{\perp}} \oplus P_{S_1}MP_{S_2} \oplus P_{S_1}MP_{S_2^{\perp}} \oplus P_{S_1^{\perp}}MP_{S_2} \\
&:= M^{\perp} \oplus M^{//} \oplus M^{//1} \oplus M^{//2} \\
&= \oplus_{j=1}^{l_1}\sigma_j^{\perp}M_j^{\perp} \oplus_{j=1}^{l_2}\sigma_j^{//}M_j^{//} \oplus_{j=1}^{l_3}\sigma_j^{//1}M_j^{//_1} \oplus_{j=1}^{l_4}\sigma_j^{//2}M_j^{//2},
\end{aligned}$$

where $\widetilde{M}_t^\lambda = \sum_{i=1}^r \sigma_i u_i v_i^\top$ with support S_1, S_2, $P_{S_1^\perp} M P_{S_2^\perp} = \sum_{i=1}^{l_1} \sigma_i^\perp u_i^\perp (v_j^\perp)^\top$ and

$$\widetilde{M}_t^\lambda + \theta M_j^\perp = \sum_{i=1}^r \sigma_i u_i v_i^\top + \theta u_j^\perp (v_j^\perp)^\top,$$

$$\widetilde{M}_t^\lambda + \theta M_j^{//} = \sum_{i=1}^r \sigma_i u_i v_i^\top + \theta u_{\alpha_j} v_{\beta_j}^\top,$$

$$\widetilde{M}_t^\lambda + \theta M_j^{//1} = \sum_{i \neq \alpha_{1,j}} \sigma_i u_i v_i^\top + \sqrt{\sigma_{\alpha_{1,j}}^2 + \theta^2} u_{\alpha_{1,j}} \left(\frac{\sigma_{\alpha_{1,j}}}{\sqrt{\sigma_{\alpha_{1,j}}^2 + \theta^2}} v_{\alpha_{1,j}} + \frac{\theta}{\sqrt{\sigma_{\alpha_{1,j}}^2 + \theta^2}} v_{\beta_{1,j}}^\perp \right)^\top,$$

$$\widetilde{M}_t^\lambda + \theta M_j^{//2} = \sum_{i \neq \alpha_{2,j}} \sigma_i u_i v_i^\top + \sqrt{\sigma_{\alpha_{2,j}}^2 + \theta^2} \left(\frac{\sigma_{\alpha_{2,j}}}{\sqrt{\sigma_{\alpha_{2,j}}^2 + \theta^2}} u_{\alpha_{2,j}} + \frac{\theta}{\sqrt{\sigma_{\alpha_{2,j}}^2 + \theta^2}} u_{\beta_{2,j}}^\perp \right) v_{\alpha_{2,j}}^\top.$$

Using the convexity of $F_t(\cdot)$, we have that for all $M, N \in \mathbb{M}$ and $B \in \partial F_t(M)$,

$$F_t(N) \geq F_t(M) + \langle B, N - M \rangle \tag{52}$$

and

$$F_t(\widetilde{M}_t^\lambda + M) - F_t(\widetilde{M}_t^\lambda) \geq \sum_{\cdot \in \mathcal{A}} \left(F_t(\widetilde{M}_t^\lambda + M^\cdot) - F_t(\widetilde{M}_t^\lambda) \right) \tag{53}$$

where $\mathcal{A} = \{\perp, //, //1, //2\}$. Because $\widetilde{M}_t^\lambda$ is the minimizer of $F_t(M)$, there exists a matrix $\widehat{B} \in \partial F_t(\widetilde{M}_t^\lambda)$ satisfies that for all $M \in \mathbb{M}$,

$$\langle \widehat{B}, \widetilde{M}_t^\lambda - M \rangle \leq 0,$$

which means that there exists

$$\widehat{V}_t \in \partial \|\widetilde{M}_t^\lambda\|_1 = \sum_{j=1}^r u_j v_j^\top + P_{S_1^\perp} \widehat{W} P_{S_2^\perp}, \quad \|\widehat{W}\|_\infty \leq 1 \tag{54}$$

satisfying that

$$2 \sum_{j=1}^T \omega_j \left(\mathcal{C}_j(\widetilde{M}_t^\lambda, \widetilde{M}_t^\lambda - M) - \left\langle \frac{1}{n_j} \sum_{i=1}^{n_j} Y_{ji} X_{ji}, \widetilde{M}_t^\lambda - M \right\rangle \right) + \lambda \langle \widehat{V}_t, \widetilde{M}_t^\lambda - M \rangle \leq 0. \tag{55}$$

Set $\varepsilon > \xi > 0$.

a. We know that for all $B_\xi \in \partial F_t(\widetilde{M}_t^\lambda + \xi M^\perp)$,

$$
\begin{aligned}
&F_t(\widetilde{M}_t^\lambda + \varepsilon M^\perp) - F_t(\widetilde{M}_t^\lambda + \xi M^\perp) \\
\geq&(\varepsilon - \xi)\langle B_\xi, M^\perp\rangle \\
=&(\varepsilon - \xi)\left[2\sum_{j=1}^{T}\omega_j\left(\mathcal{C}_j(\widetilde{M}_t^\lambda + \xi M^\perp, M^\perp) - \left\langle \frac{1}{n_j}\sum_{i=1}^{n_j} Y_{ji}X_{ji}, M^\perp\right\rangle\right) + \lambda\langle V_\xi, M^\perp\rangle\right]
\end{aligned}
$$

holds for all

$$
\begin{aligned}
&V_\xi \in \partial\|\widetilde{M}_t^\lambda + \xi M^\perp\|_1 \\
=&\left\{\sum_{j=1}^{r} u_j v_j^\top + \sum_{j=1}^{l_1} u_j^\perp (v_j^\perp)^\top + P_{(S_1\oplus_{j=1}^{l_1}u_j)^\perp} W P_{(S_2\oplus_{j=1}^{l_1}v_j)^\perp}, \|W\|_\infty \leq 1\right\}.
\end{aligned}
$$

With (54) and (55), setting $M = \widetilde{M}_t^\lambda + M^\perp$, we know that

$$
\begin{aligned}
&2\sum_{j=1}^{T}\omega_j\left(\mathcal{C}_j(\widetilde{M}_t^\lambda + \xi M^\perp, M^\perp) - \left\langle \frac{1}{n_j}\sum_{i=1}^{n_j} Y_{ji}X_{ji}, M^\perp\right\rangle\right) + \lambda\langle V_\xi, M^\perp\rangle \\
\geq&2\sum_{j=1}^{T}\omega_j\mathcal{C}_j(\xi M^\perp, M^\perp) + \lambda\langle V_\xi - \widehat{V}, M^\perp\rangle \\
=&2\xi\sum_{j=1}^{T}\omega_j\mathcal{C}_j(M^\perp, M^\perp) \\
&+\lambda\left\langle \sum_{j=1}^{l_1} u_j^\perp (v_j^\perp)^\top + P_{(S_1\oplus_{j=1}^{l_1}u_j)^\perp} W P_{(S_2\oplus_{j=1}^{l_1}v_j)^\perp} - P_{S_1^\perp}\widehat{W}P_{S_2^\perp}, \sum_{j=1}^{l_1}\sigma_j^\perp u_j^\perp (v_j^\perp)^\top\right\rangle \\
\geq&2\xi\mu\|M^\perp\|_2^2,
\end{aligned}
$$

in which we use that

$$
\begin{aligned}
&\left\langle \sum_{j=1}^{l_1} u_j^\perp (v_j^\perp)^\top + P_{(S_1\oplus_{j=1}^{l_1}u_j)^\perp} W P_{(S_2\oplus_{j=1}^{l_1}v_j)^\perp} - P_{S_1^\perp}\widehat{W}P_{S_2^\perp}, \sum_{j=1}^{l_1}\sigma_j^\perp u_j^\perp (v_j^\perp)^\top\right\rangle \\
=&\sum_{j=1}^{l_1}\sigma_j^\perp\left(1 - P_{u_j^\perp}P_{S_1^\perp}\widehat{W}P_{S_2^\perp}P_{v_j^\perp}\right) \geq 0.
\end{aligned}
$$

So we have

$$
F_t(\widetilde{M}_t^\lambda + \varepsilon M^\perp) - F_t(\widetilde{M}_t^\lambda + \xi M^\perp) \geq 2(\varepsilon - \xi)\xi\mu\|M^\perp\|_2^2,
$$

which means that

$$
F_t(\widetilde{M}_t^\lambda + \varepsilon M^\perp) - F_t(\widetilde{M}_t^\lambda) \geq \mu\|M^\perp\|_2^2\int_0^\varepsilon 2\xi d\xi = \varepsilon^2\mu\|M^\perp\|_2^2. \tag{56}
$$

b. Similarly, we know that for all $B_\xi \in \partial F_t(\widetilde{M}_t^\lambda + \xi M^{//})$

$$\begin{aligned}
&F_t(\widetilde{M}_t^\lambda + \varepsilon M^{//}) - F_t(\widetilde{M}_t^\lambda + \xi M^{//}) \\
\geq&(\varepsilon - \xi)\left[2\sum_{j=1}^{T}\omega_j\left(\mathcal{C}_j(\widetilde{M}_t^\lambda + \xi M^{//}, M^{//}) - \left\langle \frac{1}{n_j}\sum_{i=1}^{n_j} Y_{ji}X_{ji}, M^{//}\right\rangle\right) + \lambda\langle V_\xi, M^{//}\rangle\right]
\end{aligned}$$

holds for all

$$V_\xi \in \partial\|\widetilde{M}_t^\lambda + \xi M^{//}\|_1 \supset \partial\|\widetilde{M}_t^\lambda\|_1.$$

Choose $V_\xi = \widehat{V}$, so

$$2\sum_{j=1}^{T}\omega_j\left(\mathcal{C}_j(\widetilde{M}_t^\lambda + \xi M^{//}, M^{//}) - \left\langle \frac{1}{n_j}\sum_{i=1}^{n_j} Y_{ji}X_{ji}, M^{//}\right\rangle\right) + \lambda\langle V_\xi, M^{//}\rangle \geq 2\xi\mu\|M^{//}\|_2^2,$$

which means that

$$F_t(\widetilde{M}_t^\lambda + \varepsilon M^{//}) - F_t(\widetilde{M}_t^\lambda) \geq \varepsilon^2\mu\|M^{//}\|_2^2. \tag{57}$$

c. Similarly, we have that for all $B_\xi \in \partial F_t(\widetilde{M}_t^\lambda + \xi M^{//1})$

$$\begin{aligned}
&F_t(\widetilde{M}_t^\lambda + \varepsilon M^{//1}) - F_t(\widetilde{M}_t^\lambda + \xi M^{//1}) \\
\geq&(\varepsilon - \xi)\left[2\sum_{j=1}^{T}\omega_j\left(\mathcal{C}_j(\widetilde{M}_t^\lambda + \xi M^{//1}, M^{//1}) - \left\langle \frac{1}{n_j}\sum_{i=1}^{n_j} Y_{ji}X_{ji}, M^{//1}\right\rangle\right) + \lambda\langle V_\xi, M^{//1}\rangle\right]
\end{aligned}$$

holds for all

$$\begin{aligned}
&V_\xi \in \partial\|\widetilde{M}_t^\lambda + \xi M^{//1}\|_1 \\
=&\left\{\sum_{j\notin\alpha_1} u_j v_j^\top + \sum_{j=1}^{l_3} u_{\alpha_{1,j}}\widetilde{v}_j^\top + P_{S_1^\perp} W P_{(S_2/\sum_{j=1}^{l_3} v_{\alpha_{1,j}} \oplus_{j=1}^{l_3}\widetilde{v}_j)^\perp}, \|W\|_\infty \leq 1\right\}
\end{aligned}$$

with

$$\widetilde{v}_j = \frac{\sigma_{\alpha_{1,j}}}{\sqrt{\sigma_{\alpha_{1,j}}^2 + \xi^2}} v_{\alpha_{1,j}} + \frac{\xi}{\sqrt{\sigma_{\alpha_{1,j}}^2 + \xi^2}} v_{\beta_{1,j}}^\perp.$$

So we have

$$
\begin{aligned}
&2\sum_{j=1}^{T}\omega_j\left(\mathcal{C}_j(\widetilde{M}_t^{\lambda}+\xi M^{//1},M^{//1})-\left\langle\frac{1}{n_j}\sum_{i=1}^{n_j}Y_{ji}X_{ji},M^{//1}\right\rangle\right)+\lambda\langle V_\xi,M^{//1}\rangle\\
\geq&2\sum_{j=1}^{T}\omega_j\mathcal{C}_j(\xi M^{//1},M^{//1})+\lambda\langle V_\xi-\widehat{V},M^{//1}\rangle\\
=&2\xi\sum_{j=1}^{T}\omega_j\mathcal{C}_j(M^{//1},M^{//1})\\
&+\lambda\left\langle\sum_{j=1}^{l_3}\left(u_{\alpha_{1,j}}\widetilde{v}_j^T-u_{\alpha_{1,j}}v_{\alpha_{1,j}}^T\right)+P_{S_1^{\perp}}WP_{(S_2/\sum_{j=1}^{l_3}v_{\alpha_{1,j}}\oplus_{j=1}^{l_3}\widetilde{v}_j)^{\perp}}-P_{S_1^{\perp}}\widehat{W}P_{S_2^{\perp}}\ ,\right.\\
&\left.\sum_{j=1}^{l_3}\sigma_j^{//1}u_{\alpha_{1,j}}v_{\beta_{1,j}}^{\perp}\right\rangle\\
\geq&2\xi\mu\|M^{//1}\|_2^2+\lambda\sum_{j=1}^{l_3}\frac{\sigma_j^{//1}\xi}{\sqrt{\sigma_{\alpha_{1,j}}^2+\xi^2}}.
\end{aligned}
$$

Then

$$
F_t(\widetilde{M}_t^{\lambda}+\varepsilon M^{//1})-F_t(\widetilde{M}_t^{\lambda}+\xi M^{//2})\geq 2(\varepsilon-\xi)\xi\mu\|M^{//1}\|_2^2+(\varepsilon-\xi)\lambda\sum_{j=1}^{l_3}\frac{\sigma_j^{//1}\xi}{\sqrt{\sigma_{\alpha_{1,j}}^2+\xi^2}},
$$

which means that

$$
\begin{aligned}
F_t(\widetilde{M}_t^{\lambda}+\varepsilon M^{//1})-F_t(\widetilde{M}_t^{\lambda})&\geq\mu\|M^{//1}\|_2^2\int_0^{\varepsilon}2\xi d\xi+\lambda\sum_{j=1}^{l_3}\int_0^{\varepsilon}\frac{\sigma_j^{//1}\xi}{\sqrt{\sigma_{\alpha_{1,j}}^2+\xi^2}}d\xi\\
&=\varepsilon^2\mu\|M^{//1}\|_2^2+\lambda\sum_{j=1}^{l_3}\sigma_j^{//1}\left(\sqrt{\sigma_{\alpha_{1,j}}^2+\varepsilon^2}-\sigma_{\alpha_{1,j}}\right).
\end{aligned}
\tag{58}
$$

d. The result is similar to (58) as

$$
F_t(\widetilde{M}_t^{\lambda}+\varepsilon M^{//2})-F_t(\widetilde{M}_t^{\lambda})\geq\varepsilon^2\mu\|M^{//2}\|_2^2+\lambda\sum_{j=1}^{l_4}\sigma_j^{//2}\left(\sqrt{\sigma_{\alpha_{2,j}}^2+\varepsilon^2}-\sigma_{\alpha_{2,j}}\right).\tag{59}
$$

Using (53), (56), (57), (58), (59), we know that

$$\begin{aligned} F_t(\widetilde{M}_t^{\lambda}+M)-F_t(\widetilde{M}_t^{\lambda}) &\geq \sum_{\cdot\in\mathcal{A}}\left(F_t(\widetilde{M}_t^{\lambda}+M^{\cdot})-F_t(\widetilde{M}_t^{\lambda})\right) \\ &\geq \mu\sum_{\cdot\in\mathcal{A}}\|M^{\cdot}\|_2^2+\lambda\sum_{j=1}^{l_3}\sigma_j^{//1}\left(\sqrt{\sigma_{\alpha_{1,j}}^2+1}-\sigma_{\alpha_{1,j}}\right) \\ &+\lambda\sum_{j=1}^{l_4}\sigma_j^{//2}\left(\sqrt{\sigma_{\alpha_{2,j}}^2+1}-\sigma_{\alpha_{2,j}}\right) \\ &\geq \mu\sum_{\cdot\in\mathcal{A}}\|M^{\cdot}\|_2^2. \end{aligned} \tag{60}$$

Meanwhile, we can easily check that $\langle M^{\cdot}, M^{*}\rangle = 0$ hold for all $\cdot, * \in \mathcal{A},\ \cdot \neq *$, so we have

$$\|M\|_2^2 = \left\|\sum_{\cdot\in\mathcal{A}} M^{\cdot}\right\|_2^2 = \sum_{\cdot\in\mathcal{A}}\|M^{\cdot}\|_2^2. \tag{61}$$

With (51), (60), (61) and set $M = M_t^{(k)} - \widetilde{M}_t^{\lambda}$, we have that

$$\|M_t^{(k)} - \widetilde{M}_t^{\lambda}\|_2^2 \leq \frac{2L_f\gamma_t^2}{(k+1)^2\mu}.$$

Finally, with Theorem 1, the proof is finished. □

13 Proof of Corollary 4

Proof of Corollary 4: From (18), under the conditions in Corollary 3, with probability at least $1-3/(m_1+m_2)$,

$$(m_1m_2)^{-1/2}\left\|\widetilde{M}_t^{\lambda}-M_t^0\right\|_2 \leq \mathcal{C}_2\left(\frac{\sigma^{*2}(\Phi_{\mathcal{X}}\vee\Phi_{\mathcal{Y}})^2 r_t\log(m_1+m_2)}{\mu^2 m_1m_2nT}\right)^{2/5}.$$

When k_1 satisfies

$$k_1 \geq \mathcal{C}_2^{-1}\left(\frac{\sigma^{*2}(\Phi_{\mathcal{X}}\vee\Phi_{\mathcal{Y}})^2 r_t\log(m_1+m_2)}{\mu^2 m_1m_2nT}\right)^{-2/5}\left(\frac{2L_f\gamma_1^2}{\mu m_1m_2}\right)^{1/2}.$$

then

$$(m_1m_2)^{-1/2}\left(\frac{2L_f\gamma_t^2}{\mu(k_1+1)^2}\right)^{1/2} \leq \mathcal{C}_2\left(\frac{\sigma^{*2}(\Phi_{\mathcal{X}}\vee\Phi_{\mathcal{Y}})^2 r_t\log(m_1+m_2)}{\mu^2 m_1m_2nT}\right)^{2/5}$$

and thus

$$(m_1m_2)^{-1/2}\left\|M_1^{(k_1)}-M_1^0\right\|_2\leq 2\mathcal{C}_2\left(\frac{\sigma^{*2}(\Phi_{\mathcal{X}}\vee\Phi_{\mathcal{Y}})^2r_t\log(m_1+m_2)}{\mu^2m_1m_2nT}\right)^{2/5}.$$

Similarly, for $t=2,3,\cdots,T$ in the random initial, we have

$$K_0=Tk_1\geq\mathcal{C}_2^{-1}\left(\frac{\sigma^{*2}(\Phi_{\mathcal{X}}\vee\Phi_{\mathcal{Y}})^2r_t\log(m_1+m_2)}{\mu^2m_1m_2nT}\right)^{-2/5}\left(\frac{2L_f\gamma_1^2}{\mu m_1m_2}\right)^{1/2}T.$$

In the proposed initial strategy, for $t=2,3,\ldots,T$, when

$$k_t\geq\mathcal{C}_2^{-1}\left(\frac{\sigma^{*2}(\Phi_{\mathcal{X}}\vee\Phi_{\mathcal{Y}})^2r_t\log(m_1+m_2)}{\mu^2m_1m_2nT}\right)^{-2/5}\left(\frac{2L_f\gamma_t^2}{\mu m_1m_2}\right)^{1/2},$$

we have

$$\gamma_t\leq 3\mathcal{C}_2(m_1m_2)^{1/2}\left(\frac{\sigma^{*2}(\Phi_{\mathcal{X}}\vee\Phi_{\mathcal{Y}})^2r_t\log(m_1+m_2)}{\mu^2m_1m_2nT}\right)^{2/5}+\frac{D_1}{T},$$

which gives

$$(m_1m_2)^{-1/2}\left\|M_t^{(k_t)}-M_1^0\right\|_2\leq 2\mathcal{C}_2\left(\frac{\sigma^{*2}(\Phi_{\mathcal{X}}\vee\Phi_{\mathcal{Y}})^2r_t\log(m_1+m_2)}{\mu^2m_1m_2nT}\right)^{2/5}.$$

Choosing

$$\begin{aligned}K_1=\sum_{t=1}^{T}k_t&=\mathcal{C}_2^{-1}\left(\frac{\sigma^{*2}(\Phi_{\mathcal{X}}\vee\Phi_{\mathcal{Y}})^2r_t\log(m_1+m_2)}{\mu^2m_1m_2nT}\right)^{-2/5}\left(\frac{2L_f}{\mu m_1m_2}\right)^{1/2}\sum_{t=1}^{T}\gamma_t\\&\leq\mathcal{C}_2^{-1}\left(\frac{\sigma^{*2}(\Phi_{\mathcal{X}}\vee\Phi_{\mathcal{Y}})^2r_t\log(m_1+m_2)}{\mu^2m_1m_2nT}\right)^{-2/5}\left(\frac{2L_f}{\mu m_1m_2}\right)^{1/2}\times\\&\left(\gamma_1+T\left(3\mathcal{C}_2(m_1m_2)^{1/2}\left(\frac{\sigma^{*2}(\Phi_{\mathcal{X}}\vee\Phi_{\mathcal{Y}})^2r_t\log(m_1+m_2)}{\mu^2m_1m_2nT}\right)^{2/5}+\frac{D_1}{T}\right)\right)\\&=\mathcal{C}_2^{-1}\left(\frac{\sigma^{*2}(\Phi_{\mathcal{X}}\vee\Phi_{\mathcal{Y}})^2r_t\log(m_1+m_2)}{\mu^2m_1m_2nT}\right)^{-2/5}\left(\frac{2L_f}{\mu m_1m_2}\right)^{1/2}(\gamma_1+D_1)\\&+3\sqrt{2}(\frac{L_f}{\mu})^{1/2}T,\end{aligned}$$

and the proof is finished. □

For independent case, we need to obtain the error bound

$$(m_1m_2)^{-1/2}\left\|M_t^{(k_t)}-M_t^0\right\|_2\leq 2C_2\left(\frac{\sigma_*^2r_t\log(m_1+m_2)}{\mu^2m_1m_2nT}\right)^{2/5}\tag{62}$$

and the parallel result is

Corollary 9 *Under assumptions of Theorem 2, to attain* (62) *for each* $t = 1, \ldots, T$*, the total iteration step* K_0 *of random initial choice satisfies*

$$K_0 \geq \frac{1}{2} C_2^{-1} \left(\frac{\sigma_*^2 r_t \log(m_1 + m_2)}{\mu^2 m_1 m_2} \right)^{-2/5} \left(\frac{2L_f}{\mu m_1 m_2} \right)^{1/2} \gamma_1 n^{2/5} T^{7/5},$$

and total iteration step K_1 *of the proposed initial strategy satisfies*

$$K_1 \geq \frac{1}{2} C_2^{-1} \left(\frac{\sigma_*^2 r_t \log(m_1 + m_2)}{\mu^2 m_1 m_2} \right)^{-2/5} \left(\frac{2L_f}{\mu m_1 m_2} \right)^{1/2} (\gamma_1 + D_1)(nT)^{2/5} + 3\sqrt{2} \left(\frac{L_f}{\mu} \right)^{1/2} T.$$

The proof of Corollary 9 and is similar to that of Corollary 4 and hence we omit it here.

14 Additional Numerical Results

14.1 Netflix Dataset with varying the choosing of number of time intervals T

We conduct a numerical analysis using Netflix data, encompassing 1034 movies that were viewed more than 25000 times. We sample 3000 users from those whose ratings occurred more than 30 times in the dataset. The analysis focus on the first 500000 ratings recorded from October 1998 to November 2005. These ratings are randomly divided, with 80% assigned as training data and the remaining 20% as test data. To explore the impact of varying T while keeping nT constant, we consider 10 different values of T ranging from 1 to 1000 and split the observations into T time intervals in chronological order. After applying our method, results are presented in Figure 7 and Table 2.

For sufficiently large T ($T \geq 50$), the variation in T exerts a negligible influence on precision, but it results in an increase in time consumption. When T is too small ($T < 50$), merging data across a substantial temporal range into a single time point can result in a loss of temporal information, rendering the utilization of data inefficient. This observation also

corresponds to the dynamic nature inherent in the underlying low-rank matrix, signifying its dynamic essence rather than a static state.

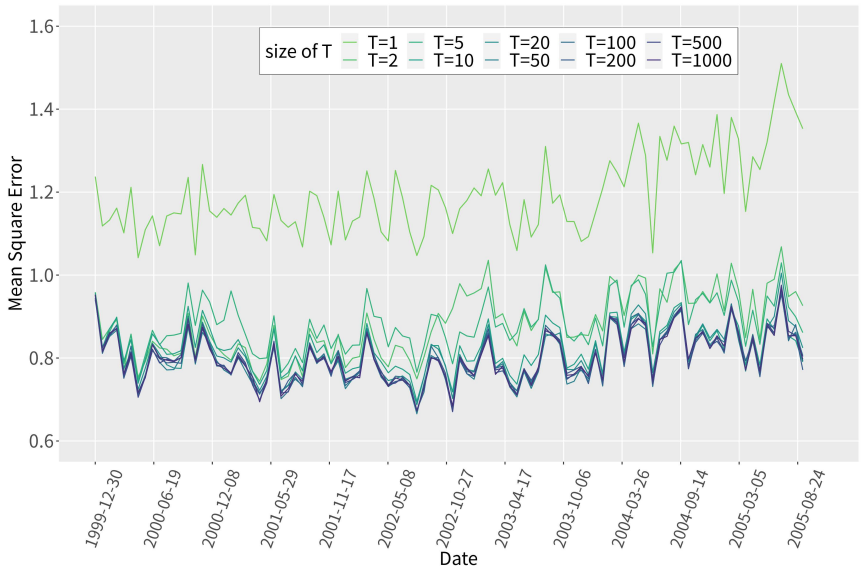


Figure 7: Mean square error for different T in each time interval.

Table 2: Average mean square error for different T.

T	1	2	5	10	20	50	100	200	500	1000
MSE*	1.194	0.881	0.886	0.825	0.812	0.800	0.803	0.802	0.801	0.802

14.2 Netflix Dataset with Link Function

In the real data experiment with Netflix dataset, each rating Y_i is a discrete integer from 1 to 5. Here we add experiment results with link function for the Netflix dataset. We assume that each user possesses a latent rating to each movie. Given the rating system's constraint of integer values ranging from 1 to 5, we consider the observed rating Y_i derived from a transformation φ on the corresponding latent rating Z_i, and assume that users typically tend to choose integer values in proximity to their latent ratings with a certain probability

as their final ratings, i.e.,

$$Y_i = \varphi(Z_i) = \begin{cases} 1 & Z_i < 0.5 \\ \lfloor Z_i + 0.5 \rfloor + \delta_i & 0.5 \leq Z_i < 5.5 \\ 5 & Z_i \geq 5.5 \end{cases}$$

where $\delta_i, i = 1, 2, \cdots, n$ follow independent Bernoulli distributions with probability $Z_i - \lfloor Z_i + 0.5 \rfloor$ to take value 1 and $\lfloor \cdot \rfloor$ is the floor operator. And we estimate M_t^0 by

$$\widetilde{M}_t^{\lambda} = \underset{M \in \mathbb{M}}{\operatorname{argmin}} \, \mathbb{E}_{\delta} \sum_{j=1}^{T} \frac{\omega_h(j-t)}{n_j} \sum_{i=1}^{n_j} \left[\mathbb{E}(\varphi(\langle M, X_{ji} \rangle))^2 - 2 Y_{ji} \varphi\left(\langle X_{ji}, M \rangle \right) \right] + \lambda \|M\|_1.$$

We followed a similar procedure and conducted new experiments with results presented in Figure 8. It is interesting to note that the inclusion of this link function performs worse. One possible reason might be that the link function may not accurately reflect the true mechanism of users' ratings. Thus, in this example, we opted to not use such a link function to illustrate the proposed method. However, this does not necessarily imply a link function is not needed in general, e.g., a more flexible form of link function by nonparametric techniques can be considered for future research.

14.3 Image Classification

An experiment was conducted utilizing the CIFAR-10 dataset to assess the efficacy of the trace regression framework in improving subsequent image-related tasks, such as classification or regression. The CIFAR-10 dataset is publicly available at https://www.cs.toronto.edu/~kriz/cifar.html. This evaluation is particularly relevant when the original dataset is plagued by incomplete information and noise. Employing a similar approach as detailed in the real-world example involving video data processing, we treated each image's individual channel as a matrix denoted by M. Subsequently, we employed the Robust Principal Component

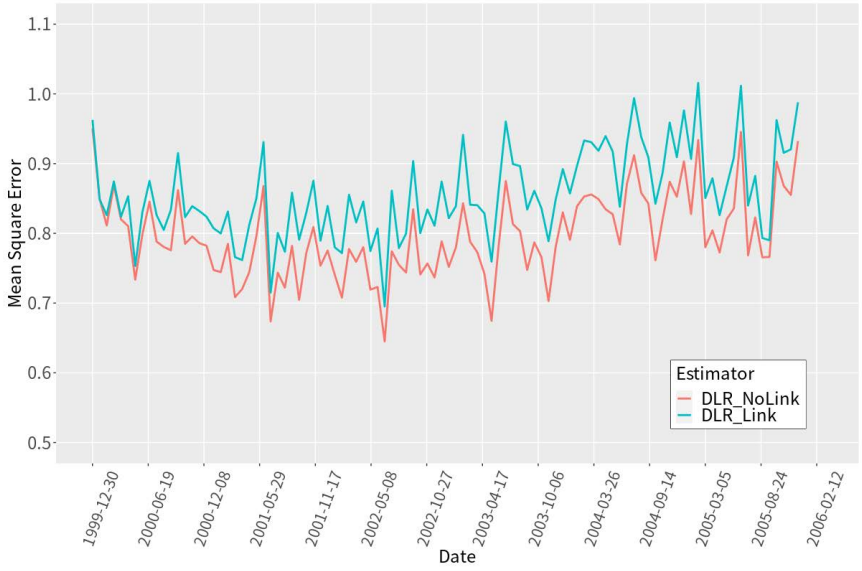


Figure 8: MSE with and without link functions

Analysis (RPCA) method to acquire the sparse (S) and low-rank (L) components, respectively. For the purpose of dataset generation, we randomly preserved a subset of elements from the low-rank component (L), subjecting these elements to perturbation using i.i.d. normal noises. The recovery process involved applying our method to retrieve the low-rank component from the saved subset of elements in L and subsequently adding the sparse component S. We compared the performance of directly using the compressed data with utilizing the data obtained after image recovery based on the trace regression framework. For the classification tasks, we utilized LeNet and ResNet18 models. The corresponding results are presented in Table 3. In the table, ρ represents the compression rate, and Acc denotes the classification accuracy in the test dataset. The findings suggest that prior image recovery before classification can lead to an enhancement in classification accuracy by approximately 3% to 5% for the aforementioned compressed datasets.

Table 3: Comparison of classification results

ρ	Network	No recovery Acc	Recovery Acc
18.5%	LeNet	50.59%	55.94%
18.5%	ResNet	74.57%	75.75%
39.1%	LeNet	52.86%	58.7%
39.1%	ResNet	76.83%	80.40%
58.6%	LeNet	54.64%	61.03%
58.6%	ResNet	77.24%	82.1%
100%	LeNet	59.09%	61.56%
100%	ResNet	80.69%	83.41%

14.4 The selection of λ and other parameters in the FISTA algorithm

We first remark that the optimal tuning parameter λ can be chosen for each time points t. The penalty term can be relaxed to depend on t, and the theoretical analysis is still valid with slight modification. In practice, based on our experience with extensive numerical studies, the universal chosen λ would not produce similarity in the low-rank structure in terms of ranks and estimated eigen-space, and can substantially reduce the computational cost associated with tuning hyper-parameters. Theoretically, considering the heterogeneous case in Assumption 7 as an example and assuming that the change of ν_t over t is not very large, i.e. there exist ν to bound ν_t such that $(1-c)\nu < \nu_t < (1+C)\nu$ with constants $c, C > 0$. With the same order assumption for sample size $n_t \asymp n$, λ and h has the optimal selection from Theorem 2 and (17) for each t by

$$h_t = C_{h,t}\left(\frac{\sigma_*^2 r_t \log(m_1+m_2)}{\nu_t^2 m_1 m_2 n T}\right)^{1/5}, \quad \lambda_t = 2C_{1,t}\sigma_*\sqrt{\frac{\log(m_1+m_2)}{nTh}}, \tag{63}$$

where $C_{h,t}$ and $C_{1,t}$ are related to M_t^0. With regularity and smoothness assumption 2, there exist $C_h, C_1 > 0$ to bound $C_{h,t}, c_{1,t}$ as their change is also not very large. So we can choose a universal bandwidth h and tuning parameter λ with the error bound for $\widetilde{M}_t^{\lambda}$ optimal up to a constant scalar.

Next, we present the detailed parameters in the FISTA algorithm of the simulation. In our proposed method, we set the tolerance (tor) to around 1×10^{-3}, which is a commonly used for the termination of optimization. For the initial time point $t = 1$, we employ the following strategy,

$$[M_1^{(0)}]_{ij} = \begin{cases} 0, & \text{if the } (i,j) \text{ element is not observed at time 1,} \\ Y_{1k}, & \text{if the } (i,j) \text{ element is observed at time 1, using the observation } Y_{1k}. \end{cases}$$

Then the output of last time $(t-1)$ is used as the initial matrix for current time t, i.e.,

$$M_t^{(0)} = M_{t-1}^{(k_{t-1})}.$$

The Lipchitz constant L_f here has an upper bound which can be calculated directly as

$$L_f = 2 \left\| \sum_{j=1}^{T} \frac{\omega_h(j-t)}{n_j} \sum_{i=1}^{n_j} (\|X_{ji}\|_2 X_{ji}) \right\|_2,$$

and the detail proof can be found in Supplementary S.6. These parameters are set in the same way for the benchmark, independent and dependent cases.

We employ a 5-fold cross-validation (CV) approach to select the tuning parameter λ in both independent and dependent settings. In our empirical findings, we observed that the results exhibit minimal differences when λ falls within a certain suitable range. To mitigate computational complexity in tuning the parameter, we suggest employing a coarse grid to estimate a rough range for λ and subsequently using a finer grid to make the final selection. When altering the matrix size, sample size, or the number of simulation points, it is not necessary to re-select λ using CV. Instead, we can rely on the theoretical relationship

$$\lambda = 2C_1\sigma_* \sqrt{\frac{\log(m_1 + m_2)}{nTh}}$$

to adjust the new λ accordingly.

14.5 Rank Changes in Simulation and Real Data Experiment

For the simulation, we show the rank of recovered matrices in independent setting with $\rho = 0.2$ and $\sigma = 1$ in Table 4 after truncating those dimensions with single values smaller than σ. For the Netflix dataset, the rank for Filter 1 is in Table 5 and for Filter 2 is in Table 6 after truncating those dimensions with single values smaller than $3\sqrt{1035}$. For the lions video dataset, the rank for low rank part L is in Table 7 after truncating those dimensions with single values smaller than $0.5\sqrt{480}$.

Table 4: Rank for Simulation in independent setting with $\rho = 0.2$ and $\sigma = 1$

T	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Rank	12	10	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9
T	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Rank	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9
T	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Rank	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9
T	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80
Rank	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9
T	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
Rank	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	11

Table 5: Rank for Netflix dataset using Filter 1

T	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Rank	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
T	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Rank	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
T	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Rank	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
T	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80
Rank	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
T	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
Rank	2	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3

Table 6: Rank for Netflix dataset using Filter 2

T	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Rank	3	3	3	3	4	4	3	3	4	4	4	4	4	3	4	4	3	3	4	3
T	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Rank	3	3	3	3	4	3	3	4	3	3	3	3	3	3	3	3	3	3	3	3
T	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Rank	4	3	3	3	3	3	4	3	3	3	3	3	3	3	4	4	3	4	3	3
T	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80
Rank	3	3	4	3	4	3	4	4	4	3	3	3	3	3	3	4	4	4	4	3
T	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
Rank	4	4	3	4	4	4	3	3	4	3	4	4	4	4	4	4	4	3	3	3

Table 7: Rank for lions video dataset

T	5	25	45	65	85
Rank (Channel Red)	6	5	6	5	6
Rank (Channel Green)	6	4	5	4	6
Rank (Channel Blue)	4	4	4	4	4